The Economics of Optional Royalties: Platform Competition, Enforcement, and Routing in NFT Markets

University of Pennsylvania
Honors Thesis in Economics

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July 15, 2026

Abstract

This paper studies how royalty enforcement policies adopted by NFT marketplaces shape platform competition, trader routing behavior, and creator revenues. Motivated by the recent divergence between marketplaces that impose mandatory royalties and those that offer optional or weak enforcement of royalties for creators, this paper develops a new three-period theoretical model with a single creator, a continuum of heterogeneous traders, and competing platforms which can choose their enforcement credibility. The model nests mandatory, optional, and zero-royalty regimes and incorporates switching costs away from a collection's minting platform. Through several analytical cases, including monopoly, symmetric duopoly, trader information and motive differences, and differing regime standoffs, the paper shows how enforcement choices interact with routing incentives. In environments dominated by speculators or with low switching frictions, platforms face downward pressures to zero enforcement credibility. When collectors or positive switching costs are sufficiently important, interior non-zero enforcement equilibria or asymmetric outcomes are shown to exist. Using Ethereum NFT trade-level data from 2022-2024, the paper additionally constructs empirical measures of royalty enforcement credibility, effective trading costs, and switching frictions. Using an estimated structural model, trader cost-sensitivity, collector valuation of enforcement, share of speculators in trading population, and price-dependent switching costs are found. Using the structural model, the paper shows that under changes to optional and mandatory regime enforcement, strengthening or weakening optional royalty compliance, and reducing switching frictions, key equilibrium changes to market share emerge. Some basic summary statistics about the data

*This paper would not be possible without the help of the following people: My peers in ECON 4900/4910 and Professor Jere Behrman for their continued advisement on my work. Professor Rakesh Vohra for introducing me to Microeconomics and teaching me the mathematical skills used in this paper. Dr. Yasser Boualam for introducing me to the digital financial markets and providing feedback on my work. Professor Jeremy Greenwood for teaching me how to numerically solve structural models. Professor Steven Matthews for introducing me to advanced microeconomic theory. Lastly, my parents Sarfaraz Lalmohammed and Karishma Sharif for always supporting my interests in firm competition. All mistakes are my own.

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1 Introduction

The recent integration of the use of Non-Fungible Tokens (NFTs) in open marketplaces has revolutionized what can be considered digtital ownership, allowing for creators to monetize any form of invention, including art, media, and other assets through programmable property rights recorded on the blockchain. With this ability, a unique property to NFTs, the ability to encode creator royalties, was added as a way to compensate artists for the sale of their work in the secondary market. Theoretically, the smart contracts in which the NFT is stored can automatically execute this payment and thus enforce it. However, practically, the enforcement of the royalties on the secondary sale of NFTs is not dependent on the permanent code in the smart contract but rather in the policies of the trading platforms which conduct the intermediate transactions between two traders/owners of an NFT. There is a widespread tension about the programmability and enforcement of the smart contracts then for NFTs and a unique opportunity to determine the optimal market structure under such circumstances.

In 2023, this issue reached a peak moment when Blur, one of the fastest growing NFT marketplaces, announced that it would no longer guarantee royalty payments on secondary sales. The decision by Blur clearly violated the immutable agreement on the contracts traded on their platform as the payments to creators promised in those contracts were no longer being made and thus the total cost to consumers dropped immediately. The previous standard for maintainng royalty rates was set by the largest NFT marketplace, OpenSea. The immediate consequence of Blur's decision was clear. Trading volume increased on the platform and retail and institutional traders alike started using the platform as the way in which they would trade NFTs. As a result, we saw what can be considered to be a race to the bottom in the enforcement of the royalties promised to creators in the smart contracts which the marketplaces provided a trading venue for. In this race though, even with non-zero royalties the trading volume that occured on OpenSea did not immediately decline. Even today, the average realized royalties, as accounted for by Dune Analytics (one of the biggest data providers on NFT markets), shows that the average realized royalty rate is still above zero. This presents an interesting outcome which suggests that some form of enforcement credibility, information asymmetry, and platform competition are working in ways to ensure partial compliance.

2 Literature Review

2.1 Theoretical Foundations

Understanding the economics of NFT royalties, particularly in an optional enforcement regime, requires background in multiple intersections subsections of the existing literature. These sections are particularly on NFT economics, two-sided platform competition, enforcement credibility, and information asymmetry. This section of the literature review will summarize the theoretical contributions which establish the conceptual basis required for understanding royalty and routing mechanisms in decentralized markets in which NFTs operate.

2.1.1 Economics of NFTs and Market Design

The foundation for modeling NFT royalties comes from an in-depth analyses of digital goods and secondary markets. Zou, Gu, and Liu (2024) develop one of the first theoretical and formal models on the operations of NFT makretplaces in which ownership and copyright are separated. The key results from their paper show that introducing a resale market alters creators optimal pricing strategy by linking the primary and secondary markets through expected resale value. They use a two stage equilibrium model linking the primary issuance of the NFTs to the secondary trading market. Their model assumes that a creator releases

NFTs that confer ownership over the asset but not the reproduction rights over that asset, which is maintained by the creators. The buyers then derive utility from both the welfare of owning the asset as well as the potential gains from reselling the NFT at a higher price than which they value it. The key innovation introduced by the paper is that the secondary market outcomes feed directly into the decisions that the creator makes in the primary market. Particularly, they find that if the expectation of resale value rises, creators would optimally lower the initial sales price in the primary market, since they understand that the bidders in the primary market can expect a profit on future resales, which benefit the creator indirectly. The paper formally derives the equilibrium conditions under which creators choose an optimal royalty rate α^*, balancing the short-term primary sale revenue with long-term gains from the movement of their created asset on the secondary market. The equilibrium royalty rate as they found it depends on the distribution of the secondary market valuations, the correlation between primary and secondary values of the NFTs, and the degree of competition among the traders of the NFTs. The authors show that overall welfare can be increased when the resale markets have a high level of liquidity and the valuations are correlated, but may reduce welfare when speculation on the future price of the asset takes priority to a genuine interest in ownership of the NFT. In the paper, the formal model introduced consists of shares of informed and uninformed traders, a marketplace on which the trading occurs, and a creator. They assume that the creators NFT is non-unique, i.e., there is no specific preferences toward one NFT compared to another and all royalties are enforced, a key integration which establishes this baseline model in which the optimal royalty rate can be found by solving the creators maximization problem with regards to profit.

Liu and Cui (2025) extend this framework by embedding NFT trading into environments with varying degrees of decentralization and interoperability. Their model focuses on a design in which there are multiple NFT marketplaces coexisting with varying enforcement capacities and degrees of interoperability, the ability to move NFT assets between markets.

Their model introduces heterogeneity in platform governance structures, particularly with some platforms having fully enforced royalties but others having them as only optional on the part of the user. The authors analyze multiple equilibrium with each of the two above cases but also that in the mixed equilibrium case which comes to be when interoperability allows NFTs to circulate across multiple platforms with different enforcement rules. The mixed equilibrium is one of the unique cases in which the authors show that it actually undermines enforcement since traders can route around high fees from marketplaces or high royalties set by the creators. However, it is clear that there exists some set of preferences by the traders since some enforcement persists when the collectors have reputation-based preferences or want to maintain high-quality NFTs. In a similar structure, Liu, Xu, and Zhu (2024) analyze the effects of business models and secondary-market structures on welfare. Their results show that the secondary markets can increase creator profits and that moderate royalties can actually increase welfare when creators internalize the resale values over time. Importantly, The paper introduces platform cost functions for royalty enforcement. In doing so, the functional form includes a component of convex enforcement cost which takes the degree of enforcement effort, a key component which counters the typical assumption of no costs to the marketplaces of effecting the royalty creators revenue stream.

2.1.2 Enforcement and Platform Competition

NFT Marketplaces function as two sided platforms connecting the creator of the NFT with traders/owners of the NFTs. In the platform competition literature, Rochet and Tirole (2003), the models of two-sided platform competition show that when agent participation on each side generates some form of network externalities, fee competition can lead to a reduction in prices or the movement toward a zero fee equilibrium. In the context of NFTs, royalty enforcement plays a role which is similar to pricing since it influences the participation of the agents and provides incentives to both sides of the market. Tunc, Cavusoglu, and

Zheng (2024) provide empirical evidence that marketplaces adjust their views on royalty enforcement and the policies they use in order to attract trading volume and this can push them down in a view similar to platform competition. It suggests that the enforcement credibility of a platform can play a role in the equilibrium allocation of the trading volume in the NFT market. Additionally, Holocomb (2025) builds on this perspective from a legal viewpoint which indicates that decentralization of NFT infrastructure makes enforcement more complicated since smart contracts cannot actually force a platform to honor royalties once trades must occur through some centralized intermediary. Enforcement credibility, in a sense, then must be modeled as an evolving decision which the platforms make with respect to the market as a whole.

Additionally, the entire conceptualization of enforcement credibility follows from the literature on theoretical relational contracting and self-enforcing agreements along with informal inforcement mechanisms, Klien and Leffler (1981) and MacLeod (2007). It is relatively straightforward to see from the existing literature and practical organization of the NFT market that the marketplaces which offer reduced royalty enforcement have the short-term gains in the form of increased fees from price-sensitive traders. However, they must balance this with the effects of creator participation in the primary market for the marketplace, where they risk losing the fees from what can be substantial primary market sales. This dynamic equilibrium for the marketplaces is similar to reputation models in repeated games for the seller, where the value of future earnings motivates opportunistic actions. This is a key distinction in the behavior and optimal decision for the NFT marketplace in choosing its enforcement regime given all creators, traders, and other marketplace decisions.

2.1.3 Network-Based Market Design

Beyond the typical representative-agent models, a recent line of theory also is concerned with the network structure underlying the market. Manea (2021) presents a general model of diffusion in networks with bottleneck links and essential intermediaries. The paper demonstrates that network design determines the surplus allocation between agents and the redundant edges can correspond to overlapping marketplaces offering similar services while bottlenecks concentrate flow and strengthen the power of the platforms or creators. Within NFT markets, platforms serve as the intermediaries connecting the creators of the NFTs and the traders/consumers of the asset. The transaction costs span across the network edges consist of marketplace fees, royalty rates, and the switching costs. Consumers would effectively want to minimize total costs incurred so they choose the path along the network edges which minimize the total fees and royalty they must pay.

When thinking about the network structure of the NFT market in the context of the existing theory literature, it becomes apparent that the operation of it can allow for many different enforcement credibility regimes to occur. Liu and Cui (2025) effectively show that in the case of interoperability between market places (adding redundant edges to the network of the NFT market), increases competition and erodes enforcement credibility. On the other hand, exclusivity or network bottlenecks by not allowing for movement of the good (the NFT) reinforce enforcement capacity for the marketplace. This means that the network-based approach can partially verify the existence of an equilibrium in which multiple enforcement regimes exist for the royalties to the creators.

2.2 Empirical Evidence from NFT Markets

2.2.1 Historical Trends in Royalties

Within the literature, it is relatively clear that the history of the NFT markets rise to its current state can be put into the context of three periods. These general periods can be considered to be the booming period of stimulated market growth in 2020 and 2021, the enforcement credibility regime drops from 2022 to 2023, and a stabilization period of slow growth which has been ongoing from 2024.

During the intial rise of the NFT market as a platform for trading in 2020 and 2021, platforms like OpenSea and Rarible automatically transfered a fixed share of resale value to creators. This was typically in the range of 5-10%. Trading activity during this time fell, for the majority of the time (greater than 80% for Ethereum-based NFTs), on one singular marketplace and enforcement regimes were clearly defined under the characteristics of the smart contracts mentioned by like ERC-721 and ERC-1155, which used the EIP-2981 interface (Qi et al., 2025). Cross-sectional analyses in Wang et al. (2024) showed that the royalties during this period acted primarily as an investment-related signal which decreased short-term resale volume but resulted in an inverted U shape when thinking about long-term value due to mixed roles of costs and quality signals. Additionally, the USD valuation of such assets varied rapidly with the exchange rate of Ether to USD varying from $730 to $4800 between January 2021 and February 2022. This is an important distinction which changes the perception of valuation to the traders for short-term changes between primary market and secondary market sales prices.

Then in late 2022, the NFT market saw the initial introduction of a drop in the enforcement regimes for royalties by Blur, and soon after, LooksRare. Particularly, these marketplaces followed optional-royalty policies to increase their trading volume. Empirically, Tunc, Cavusoglu, and Zheng (2024) show transaction volume reallocating toward venues that weakened royalty enforcement, which is clearly consistent with competitive pressures in two-sided platform competition. This followed by the period from 2024 to the present in which several marketplaces have reintroduced selective enforcement regimes. In doing so, there is the use of new technological advancements such as contract-level systems like OperatorFilterRegistry and Solana's Creator Royalty Enforcement Standard by OpenSea and Magic Eden (Qi et al., 2025). There has been a clear sustainability to the amount of royalties realized by creators then which follows from the theoretical expectations which of

Zou, Gu, and Liu (2024).

2.2.2 Empirical Findings

When it comes to the data-backed empirical findings of the existing literature, the focus is almost exclusively on the price and volume effects of NFTs, market competition, and welfare. It is notable that the specific extent to which numerical conclusions are made is limited due to the variety in NFT categories and contracts, each of which produces slightly differing effects. Using large-scale transaction datasets across multiple blockchain platforms, Sun, Jiang, and Wang (2025) find one standard deviation increase in the royalty rate significantly correlates with a 7.04% decrease in secondary market prices and a 4.8% decrease in the resale probability of NFTs. Their analysis additionally finds that NFTs with enforced royalties typically have worse investment performance compared to royalty-free tokens. In general, they state that NFT investors attach less price premium to NFTs with royalties due to information for the market value of NFTs being more symmetric under royalty rates being included. Overall, their analysis indicates price speculation for short-term holders is reduced by royalty inclusion but is somewhat irrelevant to long-term holders.

When it comes to the marketplace competition with the NFT market, Falk et al. (2025) find that for various platforms prior to the Blur drop in royalty enforcement, 65-100% of transactions were royalty enforced and that the average royalty rate was between 1% to 8% or about 0.5-9.8 ETH. A trend then becomes clear that prior to the introduction of a royalty-free policy, there was extensive platform competition with a non-zero royalty rate, even in cases where the platform fees made up to 1% of the value of the asset. This presents evidence that marketplace competition is not dependent entirely on pricing power and that some form of preferences in the context of the NFT products come to light.

2.3 Information, Behavior, and Market Outcomes

2.3.1 Informed vs. Uninformed Agents (Speculators vs. Collectors)

A unique component of the literature on NFT market economics is a focus on the behavior of the individuals in trading on the marketplaces. In the simplest case, Zou, Gu, and Liu (2024) explain explicitly in their model that there are some traders who act differently in the case of primary vs. secondary markets, what they refer to as informed and uninformed traders. Informed traders have knowledge of the primary market that allows them to come to a valuation of the NFT and submit a bid in the primary market. Uninformed traders work only in the secondary market after the NFT has been auctioned and use a valuation dependent on the primary market price. This is slightly different than the approach by Falk et al. (2025) in which the authors instead describe the behavior of various traders as being that of speculators or of collectors. They find that the NFT market is dominated by Speculators who seek to find primary market auctions undervalued such that they can resell for a profit in the secondary market. They also find that royalties enable creators to capitalize on the presence of speculators by enabling risk sharing under risk aversion , mitigate information asymmetry when speculators are better informed, and unlock price discrimination benefits. This evidence presents a clear case for the differences of behavior resulting in differing outcomes for the two types of traders and the creators. It unknowingly asks the question of whether the actions and makeup between the two different types of NFT traders can lead to a difference in the resulting equilibrium allocation of the trading volume between marketplaces.

2.3.2 Switching Costs and Routing Elasticity

The switching costs, those costs which are particular to moving an owned NFT off of one marketplace and listing it for sale on another, are a notable component of marketplace power in the NFT market. In typical two-sided platform literature, an increase in the transaction costs or frictions between exchanges can keep pricing power and allow for heterogeneity in the policies by each firm, even if the specific policies are unknown. Because switching costs are non-zero, there can exist an equilibrium in which differentiated fee structures and enforcement regimes can exist. For the NFT market, switching costs are presented both in terms of technical channels of moving the asset and social channels. On the technical side, there is the incorporation of gas fees on the Ethereum chain to change marketplaces (Etherscan) platforms, token approval fees, and loss of loyalty tokens (like that of BLUR tokens in 2023). Liu and Cui (2025) model this through their interoperability parameter, showing that limited compatabilty between platforms can allow for the sustainable royalties higher than zero. In the case that, the ability to switch, from a technical perspective, is easily accessible, traders can easily move volume towards the lowest cost platform and bypass enforcement of royalties entirely. There is also a social cost to switching marketplaces. It is possible that in doing so, traders may lose out on opportunities to participate in meaningful primary market auctions or maintain a relationship with a collection community in which they have built trust. Holcomb (2025) emphasizes that creator–collector relationships and platform-specific reputations act as soft switching frictions to traders.

Switching costs are emphasized particularly when considering them in the context of network economics. Manea (2021) clearly demonstrates that examples of bottleneck edges, those in a position where a necessary diffusion must occur, in the network graph of the market can generate stability for a marketplace even with fees. In this way, switching costs effectively serve as a countermeasure to pressures that competing marketplaces impose on one another with regards to royalty enforcement.

Following up on this idea is that of the routing elasticity, the sensitivity of the trading volume to the cross platform differences in total transaction costs inclusive of royalties. There is some clear evidence that there is some form of elasticity to enforcement regimes. The most clear of these is the 2023 case of Blurs lack of enforcement of royalties while other platforms maintained non-zero royalties and retained traders. In essence, this elasticity plays a role in order flow and thus it suggests can be modeled in a similar way to that of consumers in markets choose between differentiated products. The differences here lie between royalty policies, enforcement credibility, and switching costs. Thus, routing elasticity can be modeled similar to the product differentiation as shown in Anderson and Palma (1992) where traders choose between a number of marketplaces rahter than producsts. Formally, this means that when elasticity is high, traders instantly move away from expensive marketplaces for costs, moving towards those with zero royalties. When elasticity is low and there are preferences toward marketplaces due to their inherent characteristics, a differentiated equilibrium can exist and platforms may sustain higher enforcement credibility without losing traders.

2.4 Gaps in the Literature

Despite substantial progress in modeling NFT royalties and platform design, the existing literature lacks a unified analytical framework that integrates platform competition, enforcement credibility, and trader routing behavior within a multi-marketplace ecosystem. Current models, such as those of Zou, Gu, and Liu (2024), and Liu and Cui (2025), do one of two things. They either capture the creators optimal pricing decisions and royalty rate decisions or they consider marketplace heterogeneity in enforcement, but they do not endogenize trader flow between marketplaces as a function of relative costs and switching frictions. Similarly, theoretical studies such as Sun, Jiang, and Wang (2025) and Falk et al. (2025) provide evidence on price and volume effects of royalties, but they do not link these outcomes to equilibrium routing decisions across platforms with distinct enforcement regimes. This paper intends to fill this gap.

Additionally, no existing paper provides a formal treatment of routing elasticity and switching costs, besides some limited work on gas fees, in NFT markets despite clear behavioral evidence that traders migrate toward low-cost or royalty-free venues (Tunc, Cavusoglu, and Zheng, 2024). While Liu and Cui (2025) introduce an interoperability parameter that frames the creator problem of royalty rate selection for switching costs, their model treats trader reallocation as exogenous rather than optimizing behavior over a set of differentiated platforms. This leaves a clear gap on determining how enforcement credibility and royalty policies interact with trader preferences to shape market share. This paper is meant to addresses that gap by embedding routing elasticity and switching costs directly into a general equilibrium model of multi-marketplace NFT trading, thus linking enforcement regimes, trader behavior heterogeneity, and digital platform competition in a single, empirically testable framework.

Additionally, from an empirical perspective, there is little to no research about switching costs and the empirical observation of the typical expectation of switching given an increase in the royalty rates. While this has been done for secondary market price (Wang et al., 2024), there has been no sustained determination of the observed enforcement credibility and its effect to pricing or the probability of switching platforms for a trader. This paper intends to fill this gap by using an empirical analysis with the theoretical model to find an accurate depiction of the enforcement credibility parameter for large NFT marketplaces and the sensitivity of trading on a separate platform along with the changes to primary market auction prices.

3 General Theoretical Model

3.1 Environment and Agents

In our model, we consider a discrete time framework with three periods, similar to the existing literature. At time $t = 0$, the strategy is picked by the platform marketplaces using their trading fee, royalty regime, and enforcement credibility. At time $t = 1$, the primary market sale occurs and the creator chooses the royalty rate, the primary market price, and the minting location. This is the sale in the NFT market in which the creator receives their guaranteed income on the creation of their NFT as the primary market price is a direct transfer, minus the cost of the . In time $t = 2$, the secondary market takes place and is where traders (consumers) encounter routing shares, royalty-payment decisions, and transaction decisions while marketplaces see resulting volume. Note, each previously mentioned parameters for each time t is endogenously determined by the model.

We assume there is a single creator $\mathcal{C}$ who issues a fixed supply N of NFTs of one collection on a given blockchain. For the sake of directness, we do not consider the possibility that NFTs may be traded across chains which will allow for the extension of the theoretical model to empirical results. For most of the analysis, we will normalize $N = 1$, and all resulting quantities can be interpreted as per-unit objects. In our model, there is a finite set of NFT marketplaces $\mathcal{M}$ with $|\mathcal{M}| \geq 2$. Each marketplace $m \in M$ is a platform that intermediates NFT trades and collects fees for doing so.

The NFT market is populated by a continuum of traders of total measure 1. This means that the trader population population is modeled as a unit interval of agents, rather than as a finite set. A continuum allows us to represent heterogeneous types smoothly and ensures that aggregate shares translate directly into the corresponding masses of traders in equilibrium. No individual trader affects prices or platform conditions. As specified in the results in Zou, Gu, and Liu (2024) and Falk, Gu, Tsoukalas, and Zhang (2025), the traders differ along two independent dimensions. The information type of the traders may be informed (share $\gamma \in (0,1)$) or uninformed $(1-\gamma)$. Informed traders know the valuation distribution and correctly anticipate how platform fees, royalty enforcement, and the creator's choices affect resale value. Uninformed traders observe only broad market signals and may treat royalties or enforcement as noisy indicators of project quality. Separate from information is motive of the traders. A fraction of the traders are speculators, while the remaining portion are collectors. Speculators care primarily about trading profits. Collectors on the other hand care more about long-term consumption value, community benefits, and creator support. For specificity, if the model includes both dimensions of types of traders, let θ_{IS} be the share of speculators among informed traders and θ_{US} be the share of speculators among uninformed traders. The remaining fraction $1-\theta_{IS}$ and $1-\theta_{US}$ are the collectors in each information group.

Each potential trader (buyer) i is characterized by a pair of valuations $(v_{I,i}, v_{II,i})$ where $v_{I,i}$ is the period-1 minting value and $v_{II,i}$ is the period-2 holding/consumption value. $(v_{I,i}, v_{II,i})$ are drawn *i.i.d.* from a continuous joint distribution $F(v_I, v_{II})$. The exact form of F is not specified in this paper. For exemplification purposes, we will assume tractable cases, such as the uniform distribution used in Zou, Gu, and Liu (2024).

3.2 Platform Policies

Each marketplace $m \in \mathbb{M}$ must choose a fee, a royalty policy regime, and an enforcement credibility level. Marketplace m sets a proportional transaction fee $\beta_m \in [0, \bar{\beta}]$ charged as a fraction of the transaction price on any trade executed on platform m. By the creators choice of a royalty rate α (explained in the next section), each platform decides how it treats this royalty through a royalty policy regime. $R_m \in \{\text{mandatory,optional,zero}\}$. If in a mandatory regime, the initial agreement that the marketplace makes with the creator is that royalties must be paid on all eligible trades on m. Under the optional regime, the platform allows the buyer/seller to choose whether to honor the royalty. When in the zero regime, the platform does not route any royalty payment to the creator regardless of the choice of α (effectively sets $\alpha = 0$ for all trades on m).

As discussed in the historical context section of the literature review, even when in a mandatory or optional enforcement regime, the actual payment of the royalties can be imperfect. This may due to transactional difficulties with the NFT on the chain itself or due to the platform's immediate decision to provide additional surplus to the traders by not transferring royalty payments specified in the smart contract. In line with Sung, Jiang, and Wang (2025) estimated likelihood of royalty payment, we capture this with an enforcement credibility parameter $\kappa_m \in [0, 1]$. The direct interpretation of this parameter is it is the expected fraction of due royalties which are actually collected and passed onto the creator on platform m. It can equivalently be seen as the probability that a royalty-eligible trade is effectively enforced. Marketplaces can increase κ_m through costly enforcement (layered-security smart-contract integration, off-chain moderation, and sanctions on violators). Thus, we assume an enforcement cost function $C_m(\kappa_m)$. While realistically, this may differ between marketplaces, for illustrative purposes, and because of the homogeneous nature of the NFT minted by the creator, we will assume this cost function is consistent across marketplaces. Additionally, based on the relational and reputational penalties and computational costs discussed by Liu, Xu, and Zhu (2024) between NFT marketplaces and creators, we assume the basic properties that $C'_m(\kappa_m) > 0$ and $C''_m(\kappa_m) > 0$. This means that relaxing enforcement (reducing κ_m) can reduce strain on the NFT platform.

Letting $Q_{I,m}$ represent the primary volume (number of NFTs minted) transacted on platform m, and $Q_{II,m}$ be the secondary market volume. Let p_I and p_{II} be the primary and secondary market prices of the homogeneous NFT good. Marketplace m earns revenue on both primary and secondary trades and incurs enforcement costs. This results in a profit function defined as:

$$\Pi_m = \beta_m p_I Q_{I,m} + \delta_m \beta_m p_{II} Q_{II,m} - C_m(\kappa_m)$$

where $\delta_m \in (0, 1]$ is a discount factor capturing the present value of future secondary trades occurring at time $t = 2$.

3.3 Creator's Problem

At the beginning of the defined market game, in time $t = 1$ after the platforms have set their choices for policy and governance of the NFT, the creator chooses the primary sale price $p_I \in [0, \bar{p}]$, the royalty rate $\alpha \in [0, \bar{\alpha}]$, and the primary listing location $d \in \mathcal{M}$ which is the marketplace on which the mint occurs. If the creator would like, they may choose 2 or more marketplaces to take a share of the minting in the primary market. For most of the analysis, we will focus exclusively on primary listing in which all primary sales occur on platform d while secondary trading may occur on any platform in $\mathcal{M}$. However, in the various cases of the market structures in sections to follow, assumptions may be made on the minting location or take them to be given.

The creators payoff consists of the primary market revenue plus discounted expected royalties recieved from secondary trades. Thus the objective of the creator can be written as:

$$\Pi_C = p_I Q_{I,d}(1-\beta) + \delta_C \alpha \sum_{m \in \mathcal{M}} \kappa_m p_{II} Q_{II,m}$$

where $\delta_C \in (0, 1]$ is the creator's discount factor on the secondary market royalty revenue they expect to receive.

3.4 Trading Frictions and Effective Transaction Costs

When an NFT is chosen to be traded on platform m, the platform fee and any enforced royalties will create an effective proportional charge, but we also allow the option to the investors (traders) to switch the platform for their primary trading to that of a platform different than the primary listing. When it comes to the effective amount charged to the trader for conducting a trader with platform m at price p, the exact amount depends on royalty regime R_m, enforcement κ_m, and the trader's decision about optional royalties. Let $d_m \in \{0, 1\}$ denote whether a trader of motive type $t \in \{S, K\}$ (speculator/collector) chooses to pay the royalty on platform m when R_m = optional. Then clearly, if R_m = mandatory, then the effective charge (as a percantage of the secondary sale price) is $\tau_{m,t}(p) = \beta_m + \kappa_m \alpha$. Similarly, if R_m = optional, then $\tau_{m,t}(p) = \beta_m + d_{tm}\kappa_m \alpha$, and if R_m = zero, then we set $\alpha = 0$ for all trades on m and $\tau_{m,t}(p) = \beta_m$. We know by Liu, Xu, Zhu (2024) introduction of consumer optional royalties that speculators tend not to pay when optional while collectors may pay if they value creator support or platform privileges.

3.5 Switching Costs

Let $d \in M$ be the primary platform. If an NFT token minted on one platform is resold on platform $j \neq d$, the trader must incur a switching cost (gas, approval, lost loyalty rewards, social/community costs, etc.). We model the monetary swtiching cost as the function $k(p_{II}) \geq 0$. This is a non-decreasing function of the secondary market price p_{II}. Because we are only interested in cases of NFTs which trade on the same collection or chain, we assume such costs are similar across marketplaces. For a trade on platform m, we an define the total effective proportional cost for trader of type t as

$$\tilde{\tau}_{m,t} = \begin{cases} \tau_{m,t}(p_{II}) & \text{if } m = d \\ \tau_{m,t}(p_{II}) + k(p_{II}) & \text{if } m \neq d \end{cases}$$

3.6 Trader's decisions and Routing

We now will describe how traders choose whether to buy/sell, and if they trade in the secondary market, how they route their trades across platforms and decide whether to pay optional royalties. In the primary market, the creator offers NFTs at price p_I on platform d. A trader of type t decides whether to buy an NFT based on the period 1 valuation $v_{I,i,t}$ and their expectation of period-2 value and resale opportunities, which depend on $(\alpha, \beta_m, \kappa_m, R_m)_{min\mathcal{M}}$. If we let $U_{I,i}$ be trader i's expected net utility from buying in the primary market at p_I, then primary market demand is written as $D_I(p_I, (\alpha, \beta_m, \kappa_m, R_m)_{min\mathcal{M}}) = \{i | U_{I,i} \geq 0\}$. We assume that the creator chooses p_I so that the primary market clears so the total quantity should be equal to either all the NFTs on sale or all the demand in the primary market, which means $Q_{I,d} = \min\{N, D_I\}$. Informed traders expectations come rationally given the model parameters. They are actively aware of all costs while uninformed traders may treat the royalty rate and enforcement credibility of the platform as quality signals, which follows from Sun, Jiang, Wang (2025).

In the secondary market, valuations $v_{II,i}$ are realized. Some of the NFT owners choose to resell, while others hold. Consider an owner i contemplating resale on a platform m at secondary price p_{II}. For a speculator S, the key payoff is the expected trading profit after fees, royalties and switching costs. Thus, we define their utility in the secondary market as

$$u_{S,i} = \pi_{i,m} - \tilde{\tau}_{m,S}(p_{II})p_{II}$$

Speculators derive little intrinsic value from long-run holding and most of their utility comes from profit and cost differences. On the other hand, for a collector K, resale is motivated by consumption and community considerations. Let q_m denote an exogenous quality indicator or community appeal parameter of platform m. A collectors utility from trading on a platform m can be written as

$$u_{K,i} = v_{II,i} + \xi q_m + \phi_k \kappa_m - \rho_K p_I$$

where $\phi_k > 0$ captures the idea that collectors can prefer a higher enforcement (as a signal of support for creators and intrinsic benefits of participating on the platform), and $0 < \rho_k < 1$ indicates that collectors are less price-sensitive than speculators. Additionally, if the collectors participate in the secondary market, they incur a cost of $\tau_m p_{II}$ which they still care about regardless of turning a profit. These ideas follow from Falk et al. (2025) in which evidence is presented that royalties interact differently with speculators and genuine consumers.

Within each motive type, we assume a logit choice over the platforms, which follows from Anderson and Palma (1992), meaning that the resulting market share of type t traders can be written as a probability that each type t trades on platform m. This market share is given as:

$$s_{m,t} = \frac{\exp(\lambda_t \bar{u}_t(m))}{\sum_{l \in \mathcal{M}} \exp(\lambda_t \bar{u}_t(l))}$$

where $\bar{u}_t(m)$ is the expected utility of type t from trading on m and $\lambda_t > 0$ is a routing elasticity parameter. Typically, since speculators are more likely to be responsive to the platform fees and royalty differences, we make a logical assumption that $\lambda_S \geq \lambda_K > 0$. Thus, the aggregate secondary market share of platform m is then

$$s_m = \gamma(\theta_{IS}s_{m,IS} + (1-\theta_{IS})s_{m,IK}) + (1-\gamma)(\theta_{US}s_{m,US} + (1-\theta_{US})s_{m,UK})$$

where $s_{m,IS}$, $s_{m,US}$, $s_{m,IK}$, $s_{m,UK}$ are the market shares of uniformed speculators, uninformed collectors, informed speculators, and informed collectors respectively. Thus, the secondary volume on platform m be written as $Q_{II,m} = s_m Q_{II}$. If we let D_{II} denote the demand for the NFT in period 2 as a function of the secondary price p_{II} and the economic environment, we can state that total available supply cannot exceed the at-most N NFTs minted in the primary market, so we assume that that there is a unique secondary-market clearing price p_{II} such that $D_{II}(p_{II}, (\alpha, \beta_m, \kappa_m, R_m)_{min\mathcal{M}}) = Q_{II}$. Q_{II} is the owners of NFTs from the primary market who are willing to sell at price p_{II} in the secondary market.

3.7 Optional Royalty Payments

As a quick note for specificity to all royalty regimes, we describe the conditions for the payment of an optional royalty. Within the platform royalty regime choice $R_m =$ optional, each trader must choose whether or not to honor the royalty payments For each motive type t, we can write this choice as $d_{t,m} \in \{0, 1\}$. We assume that speculators, who care only about profits, do not pay royalties when optional (which clearly increases their profits) so that $d_{Sm} = 0$ for all m with $R_m =$ optional. Collectors however, may pay the optional royalty if their intrinsic benefit from supporting the creator outweighs the cost to do so, this can be formally modeled as

$$d_{Km} = \begin{cases} 1 & \text{if } \phi_k \kappa_m \geq \rho_k \kappa_m \alpha p_{II} \\ 0 & \text{otherwise} \end{cases}$$

4 Special Case Analysis

In this section, we consider various forms of the theoretical model applied to situations meant to illuminate a strategic component of an agent(s) behavior. The special cases are meant to isolate the core mechanisms embedded within the full model and to show how platform, creator, and trader interactions behave under different competitive and behavioral environments. Because the general model contains multiple layers of strategic interaction (platform fees, creator pricing and royalties, heterogeneous information and trader motives, routing frictions, and switching costs), solving it analytically in full generality is not very useful. Instead, each of the special cases below remove one or more dimensions of complexity from the model, such as market structure, trader types, enforcement regimes, switching frictions, or information, in order to show the underlying forces which drive the equilibrium outcomes. Together these cases will show when enforcement undercutting occurs and how it occurs, how collectors counterbalance speculators to sustain royalties, how platform competition reshapes creator incentives, how switching costs and trader-driven preferences moderate competitive pressures, and how different royalty regimes change routing behavior and equilibrium enforcement credibility choices. By deriving these simplified equilibrium, the goal is to show that the general model maintains the results previously specified in the literature on single-platform competition while creating new predictions about multi-platform interaction, trader heterogeneity, and non-zero royalty sustainability. The special cases therefore provide the theoretical foundation for the empirical strategy and the counterfactual analysis which follow in the empirical component of this paper.

4.1 Monopoly Platform, Speculators + Collectors

This subcase studies an environment in which all NFT trading occurs only on a single marketplace. Following from Zou, Gou, and Liu (2024), it serves as the baseline benchmark with no platform competition. Because there is only one platform, there is no routing competition and no strategic pressure to undercut royalty enforcement. The single monopoly platform therefore internalizes all of the trading volume created in the primary and secondary markets and then chooses its enforcement credibility solely to maximize its own revenue net of its enforcement costs. There is no worry about consumers leaving or creators switching their minting platform. This benchmark is useful because it isolates the role of the trader heterogeneity which is expanded in multi-platform competition. As described in the general model, speculators care primarily about resale profits, while collectors derive their utility from platform and creator-related attributes beyond ownership of the NFT, including the enforcement credibility of the platform.

The monopoly marketplace faces a non-trivial tradeoff. A higher enforcement credibility increases the expected royalty burden on the consumer, which makes speculative trader participation less attractive through raising total transaction costs. Alternatively, stronger enforcement can increase the attractiveness of the platform to the traders who are collectors which can interpret the signal of enforcement credibility as a signal of a legitimate marketplace. The platforms optimal enforcement choice therefore depends on whether the marginal gain in revenue from retaining or gaining collectors' trading volume outweighs the marginal loss of speculative trading volume with direct enforcement cost.

Proposition 4.1. *Consider a single NFT marketplace operating under a mandatory royalty regime with a fixed royalty rate $\alpha > 0$. There is no routing competition. A share of traders are speculators and a share are collectors. Let platform profit be*

$$\Pi_M(\kappa) = \beta \left[p_I Q_I(\kappa) + \delta_M p_{II}(\kappa) Q_{II}(\kappa) \right] - C(\kappa), \qquad \kappa \in [0, 1],$$

where $C'(\kappa) > 0$ and $C''(\kappa) > 0$.

Then:

1. If

$$\frac{d}{d\kappa}\Big(\beta\big[p_I Q_I(\kappa) + \delta_M p_{II}(\kappa) Q_{II}(\kappa)\big]\Big)\Big|_{\kappa=0} > C'(0),$$

there exists an optimal enforcement choice $\kappa^* \in (0,1]$. *If, in addition,* Π_M *is strictly concave in* κ*, then this optimizer is unique.*

2. If

$$\frac{d}{d\kappa}\Big(\beta\big[p_I Q_I(\kappa) + \delta_M p_{II}(\kappa) Q_{II}(\kappa)\big]\Big)\Big|_{\kappa=0} \leq C'(0),$$

then the monopoly optimum is $\kappa^* = 0$.

If collectors value enforcement strongly enough, the monopoly platform optimally chooses a strictly positive enforcement level. If the valuation of enforcement for collectors is weak, the platform instead sets enforcement credibility to zero. The result shows that even without platform competition, positive royalty enforcement does not occur automatically, but instead, it must be sufficiently supported by the trader-side demand for enforcement.

The interpretation of the result is as the baseline contrast with the competitive cases that follow in this section. In a monopoly marketplace, enforcement is chosen solely by balancing within-platform demand composition. In contrast, a duopoly marketplace must have the platforms worry about losing volume to rivals when enforcement raises either type of traders costs. The monopoly case therefore clarifies that it is not trader heterogeneity (i.e. the existence of speculators), which purely causes the collapse in the royalty enforcement, but instead, it is the interaction between speculative and competitive routing across platforms. Another way of interpreting this is that a monopolist platform can sustain non-zero royalties enforcement when collectors are sufficiently important, while a competitive marketplace can be forced to weaken enforcement even if there is value to it by collectors. This case is then the cleanest setting to see that trader heterogeneity alone can support positive royalty enforcement by a platform.

4.2 Symmetric Duopoly, Only Speculators, No Switching Cost

In this case, we consider a market with two competing NFT marketplaces and a population of traders only consisting of speculators. Platforms set enforcement credibility values while royalty rates and fees are symmetric and held fixed. There is no switching costs to move between one marketplace and another. Thus, traders route entirely based on total transaction costs.

Because speculative traders are purely profit driven, they naturally select the marketplace which offers the lowest total effective trading cost. Under mandatory royalties, this cost is strictly increasing in enforcement credibility. The result is that even small differences in enforcement lead to all trading volume being concentrated in the lower royalty enforcement platform. In equilibrium, this creates a discontinuous allocation of market share in which the platform with the lower enforcement should capture the entire market while the other receives no trading volume.

Proposition 4.2. *In a symmetric duopoly with marketplaces A, B with only speculators, mandatory royalties, and zero switching costs, the unique Nash equilibrium for the choice of enforcement credibility is:*

$$\kappa_A^* = \kappa_B^* = 0$$

The only equilibrium which results from these conditions for the two marketplaces is the one in which both set the enforcement credibility to zero. At this equilibrium, neither platform can profitably deviate from their chosen enforcement level as an increase in enforcement would raise the costs to the speculative traders and drive them to the other platform while decreasing enforcement would not be possible. Therefore, the competition between the marketplaces drives the enforcement to the lowest possible level.

The interpretation of this result is that it illustrates the race to the bottom in platform competition. In a case where the trading environment is dominated by speculative behavior and there is no frictions in switching platforms, positive royalty enforcement cannot be sustained. The model therefore predicts that without some collector-driven trading behavior or switching frictions, competitive pressure alone is enough to pressure marketplaces to cut out promised royalty payments entirely.

4.3 Symmetric Duopoly, Speculators + Collectors, Mandatory Royalties

Unlike the previous case, including collectors with speculators under a symmetric duopoly with a mandatory-royalty regime partially reduces the effect of reducing enforcement credibility to zero in equilibrium. Because collectors value community in the marketplace and value derived from holding the NFTs, their routing decisions between platforms depend less on the sensitivity to price differences up to a non-zero point. When royalties are mandatory, platforms cannot compete effectively by simply eliminating royalty payments altogether, but they can choose different enforcement levels for those royalties. Collectors being in the environment puts upward pressure on enforcement credibility choices which counteracts the purely profit-driven motives of the speculators. This case is meant to show how trader heterogeneity can sustain some enforcement in competition, even when speculators dominate most of the trading volume.

Proposition 4.3. *Consider a duopoly with marketplaces $m \in A, B$. Each platform chooses enforcement credibility $\kappa_m \in [0, 1]$. There is a mandatory royalty regime, the royalty rate $\alpha > 0$ is fixed, and platform fees are symmetric at β. The market consists of both speculators and collectors as traders and routing between marketplaces follows the logit structure in the general model. Collectors buy and hold for consumption only after the primary market sale.*

Let platform m's profit be given by

$$\Pi_m(\kappa_m, \kappa_{-m}) = \beta(p_I Q_{I,m}(\kappa_m, \kappa_{-m}) + \delta_M p_{II} Q_{II,m}(\kappa_m, \kappa_{-m})) - C(\kappa_m)$$

where $C'(\kappa) > 0$, $C''(\kappa) > 0$ *and assume that* $Q_{I,m}$, $Q_{II,m}$, p_I, *and* p_{II} *are continuously differentiable. Then*

1. *The marginal payoff to stronger enforcement can be decomposed into a negative speculator effect, a positive collector effect, a creator-side primary market price effect with ambiguity, a secondary market price effect with ambiguity, and a negative direct enforcement cost effect.*

2. *A symmetric interior equilibrium for enforcement credibility* $\kappa_A^* = \kappa_B^* = \kappa^* \in (0, 1)$ *exists when the marginal platform revenue gain from collector retention offsets the marginal revenue loss from speculative trading behavior and is equivalent to the marginal enforcement cost for each platform. This means that there exists such a* $\kappa^* \in (0, 1)$ *such that* $\frac{\partial \Pi_m}{\partial \kappa_m}(\kappa^*, \kappa^*) = 0$. *The sufficient condition for a positive symmetric equilibrium in royalty enforcement is* $\left.\frac{\partial \Pi_m}{\partial \kappa_m}\right|_{(0,0)} > 0$ *and* $\left.\frac{\partial \Pi_m}{\partial \kappa_m}\right|_{(1,1)} < 0$. *This results in* $\kappa^* \in (0, 1)$.

The platform problem in this case is different than the one that is with only speculators. Lowering enforcement credibility still draws in speculative trading volume, but raising enforcement creates collector volume in period 1. Platform profit is therefore a tradeoff between high frequency speculative trading volume across periods 1 and 2 in the general model and that of collector trading volume in period 1 and is dependent on the share of speculators and collectors in the market. If, collector preferences are strong and they make up a significant portion of the market population, then it is clear that a symmetric result can emerge in which both marketplaces have non-zero enforcement credibility, a clear difference between the case of only speculative traders. Thus, this result can be interpreted as showing that even in the case of competition between homogeneous platforms, the entrance of the collector type of trader can reduce the downward pressure on enforcement credibility present in the platform competition in case 2.

4.4 Symmetric Duopoly, Speculators + Collectors, Positive Switching Costs

On top of the previous case, we now additionally include positive switching costs and allow collectors to be active in the secondary market. This means that traders who obtained ownership of an NFT on a platform may continue to trade there even if it is not the directly cheapest marketplace in the secondary market. This case is meant to underscore the theory behind the idea of user lock-in that some NFT marketplaces use and the true frictions in NFT markets since switching is costly from both an administrative and technological perspective.

Proposition 4.4. *Consider a symmetric duopoly with platforms $m \in \{A, B\}$ under a mandatory royalty regime with fixed royalty rate $\alpha > 0$ and symmetric platform fee $\beta > 0$. There are both speculators and collectors, and an NFT is minted on platform d, with non-minting platform $n \neq d$. Assume that positive switching costs enter the secondary-market routing problem through an exogenous switching-cost parameter $k > 0$, paid when a trader routes a secondary-market trade away from the minting platform. Routing follows the type-specific logit structure of the general model.*

Then:

1. *At symmetric enforcement, $\kappa_d = \kappa_n = \kappa$, both trader types exhibit home-platform bias in the secondary market:*

$$s_{d,S} > \frac{1}{2}, \qquad s_{d,K} > \frac{1}{2}.$$

2. *Holding the secondary-market price fixed, the speculative routing share on the minting*

platform is strictly decreasing in own enforcement:

$$\frac{\partial s_{d,S}}{\partial \kappa_d} < 0.$$

Thus, stronger own enforcement lowers speculative routing toward the minting platform.

3. *Holding the secondary-market price fixed, the collector routing share on the minting platform satisfies*

$$\frac{\partial s_{d,K}}{\partial \kappa_d} = \lambda_K(\phi_K - \alpha p_{II})\, s_{d,K}(1 - s_{d,K}),$$

so that

$$\frac{\partial s_{d,K}}{\partial \kappa_d} > 0 \qquad \textit{if and only if} \qquad \phi_K > \alpha p_{II}.$$

Thus, collector routing rises with enforcement whenever collectors value enforcement more than the implied royalty burden in the secondary market.

4. *At a symmetric enforcement level, switching costs weaken the sensitivity of speculative routing to enforcement differences:*

$$\frac{\partial^2 s_{d,S}}{\partial \kappa_d\, \partial k} > 0.$$

Additionally, if $\phi_K < \alpha p_{II}$*, then*

$$\frac{\partial^2 s_{d,K}}{\partial \kappa_d\, \partial k} > 0,$$

so switching costs also weaken the negative collector-routing response to enforcement when collectors are primarily cost-sensitive.

5. *Given platform d's typial profit function with* $C'(\kappa_d) > 0$ *and* $C''(\kappa_d) > 0$*, suppose*

there exists a symmetric interior equilibrium $\kappa^*(k)$ *satisfying*

$$\frac{\partial \Pi_d}{\partial \kappa_d}\big(\kappa^*(k), \kappa^*(k), k\big) = 0,$$

and suppose platform profit is strictly concave in own enforcement, so that $\Pi_{\kappa\kappa} < 0$*. If the weakening of speculative undercutting from higher switching costs dominates any offsetting indirect effects through prices, primary-market quantities, and collector secondary-market volume, so that* $\Pi_{\kappa k} > 0$*, then the equilibrium enforcement level is increasing in switching costs* $\frac{d\kappa^*(k)}{dk} > 0$*.*

This result shows that positive switching costs with the heterogeneous trading structure in case 3 produces a meaningfully different quantitative result. When traders have to incur a cost to switching away from the minting platform, both collectors and speculators have a minting platform bias for the secondary market trades, even when both platforms have the same level of enforcement credibility. This matters because, if switching costs were not present, we would end up in a situation where meaningful reductions of enforcement would attract the majority of speculative trading volume in the secondary market. In contrast, when switching costs are present, the extent to which reductions in enforcement attract speculators is reduced as the size of the negative effect is being reduced as switching costs get larger.

The major implication of this case is that switching costs basically soften the rate at which the race to the bottom can occur for enforcement credibility. The profitability of decreasing enforcement credibility by the platform is reduced when traders are locked in to the minting platform through switching frictions. This allows the platform to place a higher weight on other sources of enforcement benefits to profit, like retaining collectors and supporting creators. Thus, when the weakening of speculative trader routing on platform profit from a positive change in switching costs dominates any opposing forces from prices and quantities, an increase in switching costs can raise the enforcement level chosen by the platforms.

In a total sense, this case formalizes the idea that platform competition between two platforms that are effectively the same is affected not only by differences in fee structures or royalty policies, but also by frictions in trader movement across marketplaces. If switching is costly to the traders in the market, then secondary-market order flow to the minting marketplace becomes less at-risk to be taken by the non-minting platform. As a result, royalty enforcement in the secondary market is more sustainable than in the zero-switching-cost case, and the competitive pressure to eliminate royalty-related costs for routing-based profit is reduced.

5 Empirical Model

The empirical component of this paper uses the theoretical model previously specified as a guide to quantify how royalty enforcement and switching costs affects platform routing in a setting where an marketplace (Blur) known for having less royalty compliance competes with a mandatory royalty marketplace (OpenSea). The empirical model focuses on the routing decisions at the collection-platform-time level and links those routing patterns to observable differences in total trading costs, enforcement credibility, and switching frictions. In doing so, it also considers the trader motive type composition in the existing NFT market. The empirical model does not solve for prices endogenously. Instead, it conditions on observed secondary market prices on a previously determined set of NFT transactions and uses observed routing patterns to identify a small set of the structural parameters governing the cost sensitivity, switching costs, and how much traders value enforcement.

5.1 Empirical Data and Simplification

Empirically, I work with Ethereum chain NFT trade-level data from 2022-2024 from Dune Analytics (a data provider on chain-level transactions). The data aggregates all trades in which the executing platform was OpenSea or Blur and the collection created was also on OpenSea or Blur. The idea is that this narrows focus to the strategic interaction between the two marketplaces without having to consider outside forces. This is robust considering that the two marketplaces together makeup the vast majority of NFT trades since 2022 and so any other marketplaces have a limited effect on the strategic decisions between the two. The data focuses on secondary sales with platform labels of optional vs. mandatory, the trade price $p_{II,i}$, the royalty paid amount, the platform fee amount, collection address, token ID, timestamp. It also contains the collection level information with the stated royalty rate α_C and the primary mint platform d_c (inferred to be the first . The empirical model will consider Blur to be given as platform B and OpenSea to be written as platform O. The primary question to answer is how do differences in $\kappa_{B,c,t}, \kappa_{O,c,t}$ and in total trading cost $\tau_{B,c,t}, \tau_{O,c,t}$ show up in routing shares and switching behavior in secondary markets.

5.2 Constructed Measures

To connect the model to the data, we construct a small set of key variables at the platform-collection-time level. First, for the enforcement credibility parameter, for each platform $m \in \{B, O\}$, collection c, and period t (which is one week), we define:

$$\kappa_{m,c,t} = \frac{\text{Sum of Total Royalty paid on trades of } c \text{ on } m \text{ in } t}{\alpha_c(\text{total price volume on those trades})}$$

For total effective trading cost, $\tau_{m,c,t}$, using the platforms fee rate and royalty rate of each collection, we can construct the following measure:

$$\tau_{m,c,t} = \beta_{m,c,t} + \alpha_c \kappa_{m,c,t}$$

For market shares, using the volume of each individual collection that is traded in the market over the time t, we can compute the platform-secondary market shares as

$$s_{m,c,t} = \frac{Q_{II,m,c,t}}{\sum_{j \in \{B,O\}} \sum_d Q_{II,j,d,t}}$$

Additionally, in order to capture the switching component in the secondary market, an indicator is defined for whether a trade is executed off the minting platform. We compute for each c, t, the fraction of the secondary trades that stay on the minting platform or switch to the other marketplace.

5.3 Data Cleaning

A data cleaning procedure was also implemented after calculating the constructed measures. After restricting the data to Blur and OpenSea observations within the estimation time-frame, all the variables in the estimation process were required to be numeric, finite, and non-missing. Cleaning was done at the collection-week pair level. This means that each market had to contain at least one valid Blur row and one valid OpenSea row, so as to represent the market accurately between the two individual marketplaces. To exclude markets with extreme values or those whose liquidity did not adequately meet the criteria to be considered between duopoly platform marketplaces, a series of filtering tests were applied. To exclude thin and potentially unstable markets, collection–weeks were required to have at least 10 total trades and at least 2 ETH of total collection-week volume, while each platform-level observation was required to have at least 2 trades and at least 0.10 ETH of platform-specific secondary volume. Royalty rates were restricted to the range 0.5–10.5 percent, enforcement credibility was clipped to the interval $[0, 1]$, and observations with nonpositive prices or volumes were removed. To limit the influence of extreme outliers, price, volume, and trade-count variables were cut off at the 0.5th and 99.5th percentiles, and share variables were clipped away from zero and one using a floor of 10^{-6} so as to ensure that there was no monopoly in any of the markets being analyzed and firm competition existed. The final step to the cleaning process, which has a strong influence on the resulting parameter estimation, was that if any of the validity tests for the data failed for one of the marketplaces during that collection-week, the entire collection-week observation for both marketplaces was removed so as to maintain only two-sided secondary markets with accurate data. The result was $n = 14,648$ unique rows of data with $7,324$ paired market observations for estimation on. Some basic summary statistics of the data can be found in Table 1.

Table 1: Summary Statistics for the Medium-Cleaned Paired Dataset

Variable	Mean	SD	P25	Median	P75	Min / Max
Platform-week trades	66.578	155.900	7.000	19.000	55.000	2.000 / 1299.080
Platform-week secondary volume (ETH)	27.379	78.895	1.699	4.586	15.990	0.153 / 698.947
Average trade price (ETH)	0.692	1.525	0.109	0.246	0.603	0.007 / 14.300
Royalty rate (pct)	5.276	1.778	5.000	5.000	5.000	0.500 / 10.500
Platform fee rate (pct)	0.534	0.604	0.000	0.318	0.851	0.000 / 2.587
Enforcement credibility	0.375	0.343	0.100	0.227	0.592	0.000 / 1.000
Effective total cost rate (pct)	2.536	2.499	0.603	1.500	3.999	0.000 / 12.500
Volume share	0.500	0.334	0.159	0.500	0.841	0.001 / 0.999
Trade share	0.501	0.330	0.200	0.500	0.800	0.000 / 1.000
Switching share (volume)	0.322	0.283	0.097	0.211	0.489	0.003 / 0.999
Switching share (trades)	0.326	0.280	0.100	0.200	0.500	0.000 / 1.000
Collection-week trades	138.031	297.311	27.000	52.000	115.000	10.000 / 2571.440
Collection-week volume (ETH)	57.995	146.681	4.895	12.695	42.618	2.031 / 1281.285

5.4 Empirical Structural Model

This section develops the empirical structural model used to study platform competition between Blur and OpenSea from 2022-2024 in secondary NFT markets. The objective of the model is to explain how trading activity is allocated across platforms as a function of platform fees, royalty enforcement, and switching frictions associated with trading away from the mint platform.

5.4.1 Switching frictions and mint structure

A central feature of the model, as motivated by the theoretical examples, is that trading away from the minting platform is costly to the trader. Let $1\{\textbf{B is mint}\}$ be an indicator variable noting for any collection that Blur is the minting platform. As in the theoretical model, off-mint trading is modeled as subject to a switching cost that depends on the observed secondary market price of the collection. Let $p_{O,c,t}$ and $p_{B,c,t}$ represent the average trade price prices on OpenSea and Blur at the collection-time level. Then define the standardized log-prices as

$$z_{B,c,t} = \frac{\log p_{B,c,t} - \overline{\log p_B}}{\sigma(\log p_B)}, \qquad z_{O,c,t} = \frac{\log p_{O,c,t} - \overline{\log p_O}}{\sigma(\log p_O)}.$$

Then these standardized log-prices enter the switching cost function. For OpenSea and Blur respectively, the off-mint switching cost functions are represented as

$$k_B(p_{B,c,t}) = k_0 + k_1 z_{B,c,t}, \qquad k_O(p_{O,c,t}) = k_0 + k_1 z_{O,c,t}$$

which are linear functions of the standardized log price. The k_0 captures the fixed cost to switching between the two platforms. The parameter k_1 allows the off-mint switching frictions to vary systematically with price. The explanation for the use of the standardized log-prices rather than raw prices is due to the heavy skewness of NFT price distributions and it gives the slope of the switching cost function an interpretable definition in terms of above-average and below-average price markets. Additionally, as in the theoretical model, switching costs enter the total costs to the trader when they trade on the non-mint platform. Thus, the switching-adjusted total cost functions as percentage of the secondary market price are

$$\tilde{\tau}_{B,c,t} = \tau_{B,c,t} + 1\{\textbf{B is not mint}\}k_B(p_{B,c,t})$$

for Blur-based trades and

$$\tilde{\tau}_{O,c,t} = \tau_{O,c,t} + 1\{\textbf{B is mint}\}k_O(p_{O,c,t})$$

for OpenSea-based trades. Then the difference in total effective costs to the traders can be represented as

$$\Delta\tilde{\tau}_{c,t} = \tilde{\tau}_{B,c,t} - \tilde{\tau}_{O,c,t}.$$

5.4.2 Trader heterogeneity

Like the theoretical model, the empirical version of the model assumes that there are two types of traders, speculators and collectors. Speculators are profit-seeking traders and thus only care about the total costs they will incur on the platform. Collectors, as they are trading in the secondary market, care about total costs incurred but also the enforcement of royalties for creators. Let $\Delta\kappa_{c,t} = \kappa_{B,c,t} - \kappa_{O,c,t}$ represent the difference between the observed enforcement of Blur and OpenSea in collection c at time t. Then the latent utility index for Choosing Blur rather than OpenSea for a Speculator is given as

$$V_{S,c,t} = \delta_B - \lambda\Delta\tilde{\tau}_{c,t}$$

and for collectors it is

$$V_{K,c,t} = \delta_B - \lambda\Delta\tilde{\tau}_{c,t} + \phi_K\Delta\kappa_{c,t}$$

The parameter δ_B represents the baseline Blur preference by both collectors and speculators that cannot be explained either by the difference in enforcement of royalties or the difference in total costs incurred. As in the theory presented, the λ represents the cost sensitivity of collectors and speculators to costs, and the model assumes that the two are going to be equally sensitive to costs but route differently due to differences in preferences for enforcement of royalties in ϕ_K for the collectors. This decision makes the interpretation of the collector channel clear where the primary difference between collectors and speculators is their valuation of enforcement.

5.4.3 Platform choice probabilities

Given the latent utility indexes for the collectors and speculators, the empirical model assumes that a logit routing structure synonymous to the theoretical model. Thus, the share of speculators and collectors on Blur can be represented as

$$s^S_{B,c,t} = \frac{1}{1+\exp(-V_{S,c,t})}, \qquad s^K_{B,c,t} = \frac{1}{1+\exp(-V_{K,c,t})}$$

OpenSea's trader-type specific shares are simply the complements of Blur's which are

$$s^S_{O,c,t} = 1 - s^S_{B,c,t}, \qquad s^K_{O,c,t} = 1 - s^K_{B,c,t}$$

At the market level, the model makes a strong assumption that there is a consistent share of speculators $\gamma \in (0,1)$. Then the aggregate predicted market share for blur is represented as

$$\hat{s}_{B,c,t} = \gamma s^S_{B,c,t} + (1-\gamma) s^K_{B,c,t}$$

while the predicted market share for OpenSea is

$$\hat{s}_{O,c,t} = 1 - \hat{s}_{B,c,t}.$$

This mixture model structure, following the theoretical structure, allows the model to rationalize the market shares as the weighted outcome of two populations with different motivations for platform choice.

5.4.4 Market outcomes used in estimation

The estimator for the model is based around two observable outcomes which are the share of secondary volume difference (referred to as the share gap) between the two marketplaces and the proportion of secondary market volume which switches to the competing platform from the minting platform (referred to as the switching share). Formally, the share gap is defined as $s_{B,c,t} - s_{O,c,t}$. Since there are only two platforms, this can be directly written as

$$s_{B,c,t} - s_{O,c,t} = s_{B,c,t} - (1 - s_{B,c,t}) = 2s_{B,c,t} - 1$$

Thus, the model-implied share gap is calculated as

$$\widehat{\text{share gap}}_{c,t} = 2\hat{s}_{B,c,t} - 1.$$

The interpretation of this metric is that a value of zero represents a perfect split of the secondary volume between the two platforms while a positive value emphasizes a higher share towards Blur and a negative value indicates OpenSea has a greater share.

The second outcome of interest is the switching share which is formally defined in the model as

$$\widehat{\text{switch}}_{c,t} = \begin{cases} 1 - \hat{s}_{B,c,t} & \text{if Blur is mint} \\ \\ \hat{s}_{B,c,t} & \text{if Blur is mint} \end{cases}$$

This is the model counterpart to the empirical amount of secondary volume conducted in the non-minting marketplace which is included in the dataset. It is not a dynamic notion of traders moving across platforms over time, but rather, it is the share of collection-week trading routed away from the mint platform.

5.5 Moments

The main principle in choosing the moments for use with the empirical model was to use moments that are easily economically interpretable and directly connected to the core mechanisms of the model of interest. The moments chosen focus on reproducing the relationship between three broad empirical patterns. First, they match how market share varies with cross-platform differences in effective cost. Second, they reproduce how aggregate market share varies with cross-platform differences in enforcement credibility. Finally, they focus on the amount of off-mint trading (or switching trading-share of volume), particularly in markets where mint structure and price-related switching frictions are likely to matter. These three empirical dimensions connect into the three most important parameter groups in the model which are the cost-sensitivity parameter λ, the enforcement-preference parameter ϕ_K, and the switching parameters k_0 and k_1.

The first category of moments chosen for the estimation are share gap moments. Specifically, the moments are the binned means of the share gap. The first share gap moment groups markets by collection-level royalty rate α_c which falls into one of the following bins: $[0, 1), [1, 2.5), [2.5, 5), [5, 7.5), [7.5, 10), [10, 15)$. For each royalty bin, the moment is the mean observed share gap in each bin which is written as $\mathbb{E}[s_B - s_O|\alpha_c \in$

bin]. This moment is included because the role of royalties is central to the paper's claim that royalty intensity affects the magnitude of effective cost differences and the corresponding importance of enforcement. If the model is to capture the economics of creator compensation and platform competition, it must reproduce how market shares vary across low-royalty and high-royalty collections. The second share gap groups markets by the mean cross-platform enforcement difference $\Delta\kappa$ which falls into the following bins: $[-1, -0.75), [-0.75, -0.5), [-0.5, -0.25), [-0.25, -0.10), [-0.10, 0), [0, 0.10), [0.10, 0.25), [0.25, 0.5), [0.5, 0.7$ Then the moment is formally defined as $\mathbb{E}[s_B - s_O | \Delta\kappa \in \text{bin}]$. This moment is particularly important for identifying ϕ_K because if stronger enforcement on OpenSea is associated with systematically higher OpenSea shares in the data, then the model must reproduce that pattern using the enforcement-preference channel. The final share gap moment is the one which groups markets by the mean effective total cost difference $\Delta\tau$ which falls into the following bins: $[-7.5, -5), [-5, -2.5), [-2.5, -1), [-1, -0.5), [-0.5, 0), [0, 0.5), [0.5, 1), [1, 2.5), [2.5, 5), [5, 7.5)$. It is formally defined as $\mathbb{E}[s_B - s_O | \Delta\tau \in \text{bin}]$. This moment is particularly important for identifying λ since both traders don't respond well to rising costs, so this tells the model how much the market share of Blur should respond to total incurred trader expense differences between it and OpenSea.

The second category of moments chosen for the estimation are switching moments. The primary moment in this category is the one which groups markets by effective total cost difference and is thus defined as $\mathbb{E}[\text{switch} | \Delta\tau \in \text{bin}]$. This moment is included because the switching-cost function enters utility through the effective cost of trading on the off-mint platform. If switching frictions are important in the data, then observed off-mint trading should vary systematically with relative costs. This moment therefore provides the main identification for the switching-cost parameters k_0 and k_1.

The third category of moments chosen for estimation are mean moments. Unlike the binned moments, which match on multiple values, the mean moments are simply defined as

$\mathbb{E}[s_B]$ and $\mathbb{E}$[switch] which are the mean share of Blur in all markets and the mean switching share in all markets respectively. These moments directly help anchor the model to not only understand the general patterns in the data but also the absolute value of Blur's dominance and the switching share off the mint platform. They are useful in identifying δ_B.

The fourth category of moments chosen for estimation are subgroup moments. As part of the analysis, we consider the key patterns within subgroups of the data we are interested in. The first of these groups is the cases when Blur is the mint platform and when it is not. In each case, we define a moment that is the mean of the Blur share and the mean of the switching share. The second subgroup category we consider is price groups. We calculate the average of the log-price for both marketplaces and classify a collection-week as high-price if it falls above the sample median and low-price if it falls below. Again, for each price group, we define a moment that is the mean of the Blur share and the mean of the switching share. The third subgroup category we consider is τ-parity and non-τ-parity which is effectively defined as when the difference in total effective costs between the two marketplaces is very low and near zero, defined as $|\tau_B - \tau_O| \leq 0.50$. Thus, if the total effective costs are very similar the market falls into the τ-parity group, and if not, then it is non-τ-parity. We define moments for the mean Blur share and mean switching for the τ-parity group. The last subgroup category is κ-parity and non-κ-parity which is defined as when the difference in enforcement credibility between the two platforms is very low and near zero, defined as $|\kappa_B - \kappa_O| \leq 0.05$. Thus, if enforcement levels are very similar the market falls into the κ-parity group, and if not, then it is non-κ-parity. We define a moment for the mean share of Blur for the κ-parity group.

The final category of moments chosen for estimation are slope moments. Specifically, we define the moments of the slope of the share gap with respect to effective cost difference, the slope of switching with respect to effective cost difference, and the slope of the share gap with respect to enforcement difference. These slopes are computed as covariance-overvariance coefficients and thus defined as

$$\text{slope}(\Delta\tau, s_B - s_O) = \frac{\text{Cov}(\Delta\tau, s_B - s_O)}{\text{Var}(\Delta\tau)}, \qquad \text{slope}(\Delta\tau, \text{switch}) = \frac{\text{Cov}(\Delta\tau, \text{switch})}{\text{Var}(\Delta\tau)}$$

$$\text{slope}(\Delta\kappa, s_B - s_O) = \frac{\text{Cov}(\Delta\kappa, s_B - s_O)}{\text{Var}(\Delta\kappa)}$$

The slope of the share gap with respect to $\Delta\tau$ reinforces the identification of λ. Even if the model matches binned means reasonably well, it must also reproduce the overall directional relationship between difference in costs to differences in market shares. Similarly, the slope of switching with respect to $\Delta\tau$ provides more information about the switching-cost function, and the slope of the share gap with respect to $\delta\kappa$ helps more accurately identify ϕ_K. Thus, this category of moments helps reproduce broad patterns in the data without relying on binned means.

5.6 Estimation

With the above structure, the empirical strategy is to first compute the empirical moments from the data. This follows directly from the previous section's descriptions. Then we simulate a $\theta = (\gamma, \lambda, \delta_B, k_0, k_1, \phi_K)$. We plug in the observed $p_{c,t}, \tau_{O,c,t}, \tau_{B,c,t}, \kappa_{O,c,t}, \kappa_{B,c,t}$ into the logit share equation from the empirical model and then compute the implied market shares of the model using the parameters, then aggregate these into the model moments $M^{model}(\theta)$. We can then estimate the structural model by choosing $\hat{\theta}$ to minimize the distance between the empirical moments and model moments, which is given as:

$$\hat{\theta} = \arg\min_{\theta}(M^{data} - M^{model}(\theta))W(M^{data} - M^{model}(\theta))$$

5.7 Estimation Results and Consistency

This section presents the results of the structural estimation and evaluates the extent to which the model is able to reproduce the empirical patterns observed in the data. We assess whether the estimated model provides a consistent aggregate representation of platform competition between Blur and OpenSea. In particular, we argue that the model can rationalize the variation in market shares, switching behavior, and enforcement-related differences that appear in the data using the structural mechanisms introduced in the previous sections.

Table 2: Parameter Estimates and Bootstrap Confidence Intervals

Parameter	Estimate	Boot Mean	Boot SD	2.5%	50%	97.5%
δ_B	0.9166	0.9216	0.0320	0.8674	0.9218	0.9790
λ	0.1934	0.1798	0.0405	0.1158	0.1755	0.2723
ϕ_K	3.3751	3.5527	2.4823	0.5127	3.6486	8.0000
k_0	1.6105	1.6709	0.5643	1.1393	1.6054	2.3290
k_1	5.0000	4.8334	0.4247	3.5338	5.0000	5.0000
γ	0.8050	0.7417	0.2392	0.0052	0.8370	0.9175

Table 2 reports the estimates together with bootstrap summary statistics and percentile confidence intervals. Each of the parameter estimates are important to discuss. First, the estimated Blur intercept, δ_B, is positive and relatively stable, with an estimate of 0.917 and a 95 percent bootstrap interval of approximately [0.867, 0.979]. This indicates that, even after controlling for effective cost differences, enforcement credibility, and switching frictions, the model finds that Blur has a positive baseline preference from traders relative to OpenSea.

Next, the estimate for the cost sensitivity parameter, λ, is clearly positive and again relatively stable at 0.193. Since we know that the utility indices of the traders were defined in terms of subtracting this parameter times the difference in effective total costs by percentage, this indicates that the model finds that traders are generally against rising total costs in terms of preferences, lining up well with the theory. Additionally, the estimate of the collectors preference for enforcement, ϕ_K, is positive at 3.375. We know that this enters the utility for collectors as the product with the difference in enforcement and thus this signals that collectors do positively prefer enforcement according to the model. However, as seen by the wider interval, the exact value of this parameter is less directly pinned down by the structural model.

Additionally, we can consider the estimates for the switching cost function. Both k_0, the fixed cost to switching platforms estimated at 1.611, and k_1, the marginal cost estimated at 5, are positive in the entire 95% bootstrap interval. Clearly, by width of the intervals, k_0 is nailed down pretty well and the variable cost, by its magnitude, strongly signals a steep relationship between NFT price and the cost of trading on the non-mint platform. Finally, the estimated share of speculators in the market, γ, is estimated at 0.805 or about 80.5%. This is relatively high, and the model does struggle to nail down this share exactly which is seen by the very wide bootstrap intervals. Regardless, this estimate indicates that a substantial fraction of platform routing behavior is explained by the type whose platform choice is governed solely by effective cost considerations.

Beyond the parameter estimates, we can additionally consider how well the model fits to the overall distribution of the market share and switching data.

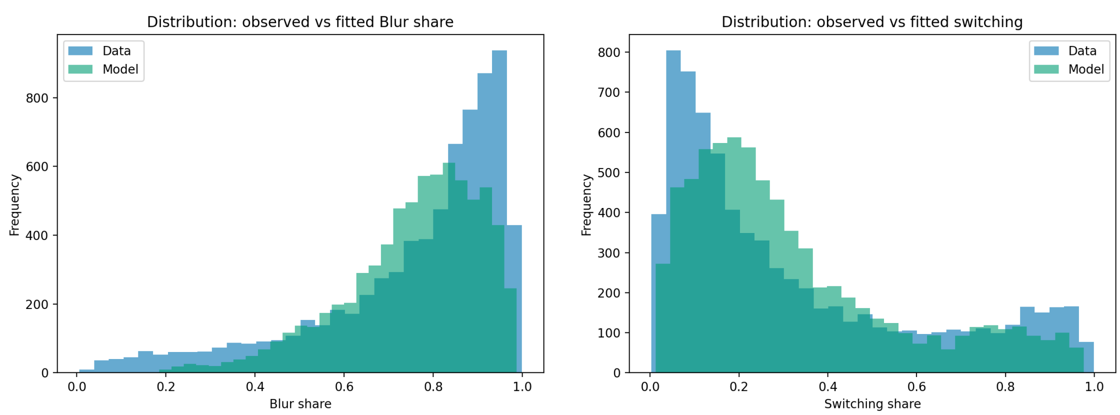

Figure 1: Distribution of Blur Share and Switching Share

Figure 1 compares the observed and fitted distributions of Blur share and switching share. The left panel shows that the model recreates the overall right-skewed distribution of Blur market share since in both the data and the fitted model, a large mass of collection-week markets is concentrated at relatively high Blur shares. However, the fitted distribution is smoother and more concentrated than the empirical one, which has a greater concentration of values at the extremes, indicating that the model captures the Blur dominance but does not completely reproduce the exact distribution of market-level shares. The right panel compares the observed and fitted distributions of switching share. The model again captures the basic shape of the data, including the concentration of markets at relatively low switching levels and the long right tail. It is still smoother than the true data.

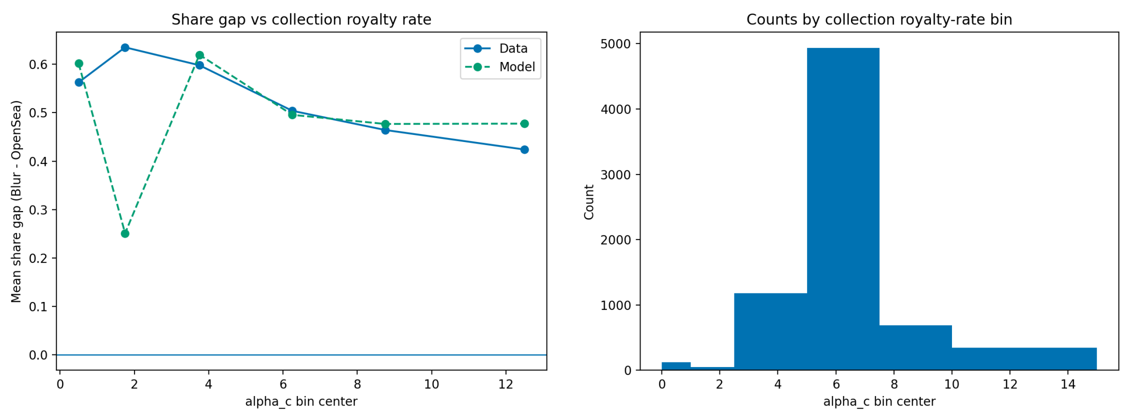

Figure 2: Fit of Share Gap vs. Collection Royalty Rate

Figure 2 is focused on examining the fit of the share gap against the binned collection-level royalty rate. The left panel shows the observed and fitted values of this moment and the right shows the frequency for the royalty-rates for context. Clearly, in cases in which there are a significant number of markets, the model picks up on the general tendency of the share gap to drop as the royalty rate for the collection increases. Overall, this figure supports that the model is able to capture the relationship between the collection-stated royalties and platform allocation, especially in the empirically important mid-range and high-ranges of the royalty distribution.

Figure 3 is focused on the fit of the moment of the share gap to binned means of the difference in enforcement between Blur and OpenSea. The observed data shows that the share gap is relatively flat or slightly declining in $\Delta\kappa$ which could be support for a higher

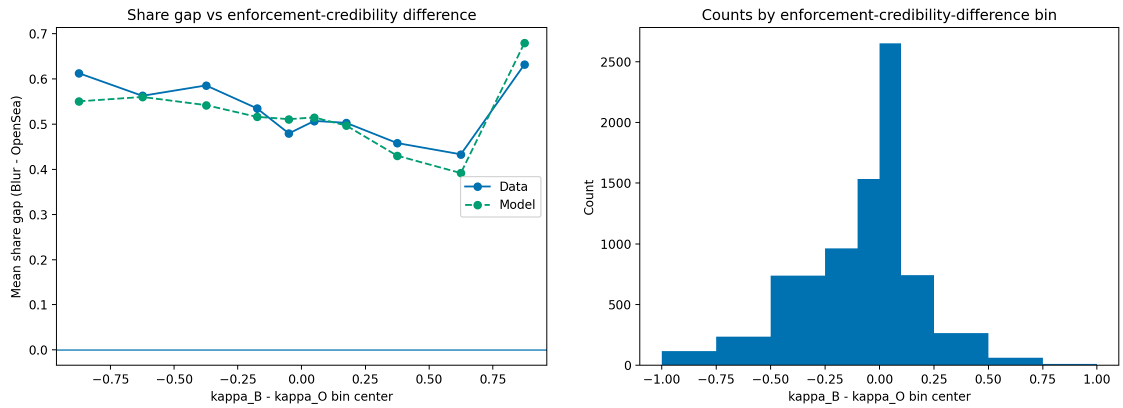

Figure 3: Fit of Share Gap vs. $\Delta\kappa$

share of speculator in the market. This is because in general, the data is saying that higher enforcement from Blur, generally results in a stronger market share for OpenSea. The strongest positive movement appears only in the far right extreme, where Blur's enforcement advantage is largest and thus, it is likely the point at which collectors dominate. In general, the model does a strong job of tracking the exact pattern of the data in this moment, including the results in the bin in which Blur's enforcement dominates.

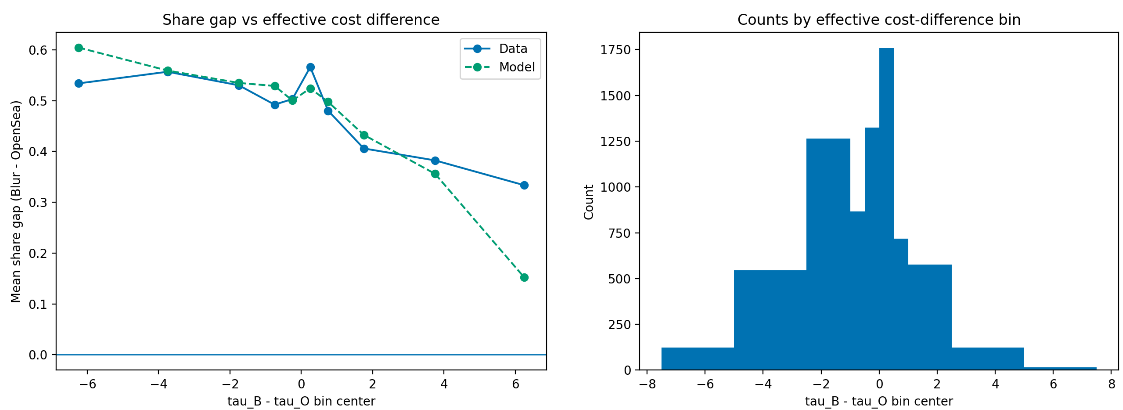

Figure 4: Fit of Share Gap vs. $\Delta\tau$

Figure 4 is focused on the fit of the moment of the share gap to binned means of the difference in total effective costs between Blur and OpenSea. The interpretation of the data side of this moment is relatively straightforward. It points directly to a negative relationship in that increasing total effective costs by either platform is indicative of a lower market share.

The model does a strong job of following this same pattern for all the bins in which there are enough markets. However, the extreme, where there are little markets ($\Delta\tau > 6$ bins), have a higher than expected reduction in relative market share for Blur under the prediction by the model.

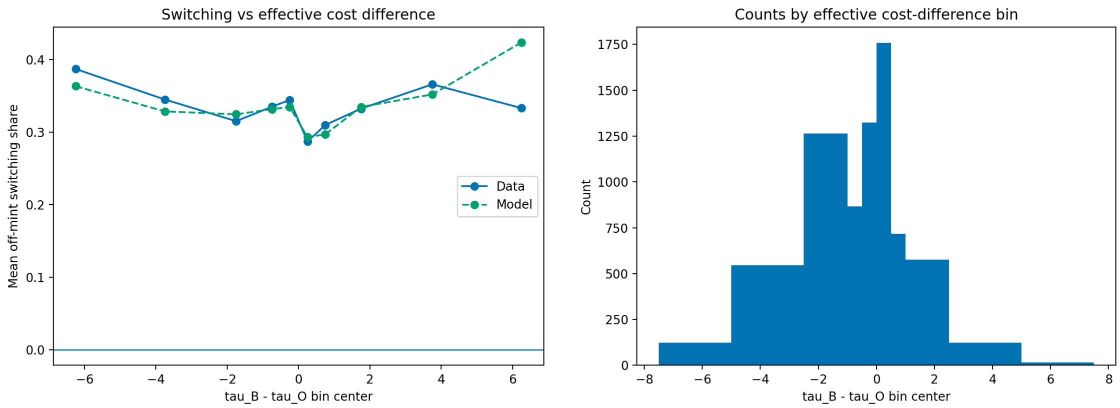

Figure 5: Fit of Switching Share vs. $\Delta\tau$

Figure 5 is focused on the fit between the switching share of the secondary market volume and the effective cost difference between the two platforms. As expected, we observe a u-shaped result in the data in which higher relative costs by either platform result in a higher off-mint switching share. Thus, if Blur is the minting platform and decides to increase costs relative to OpenSea, then it is associated with an increase in the secondary market switching share to OpenSea. The model's predicted values match this shape and pattern well, except for those bins in which there is very little data available.

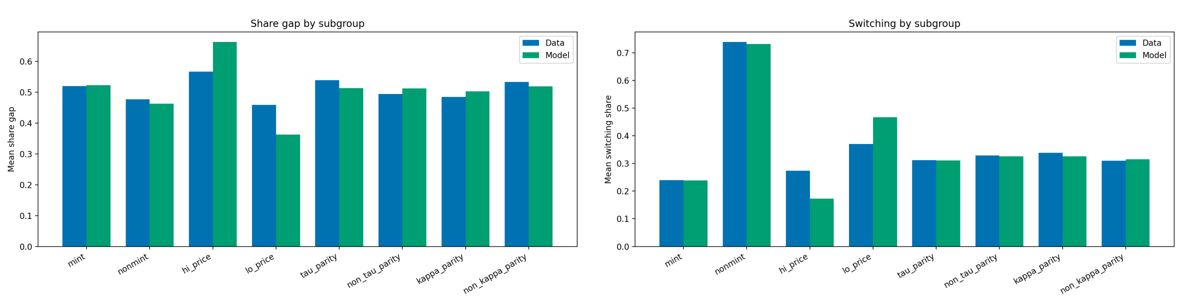

Figure 6: Fit of Subgroup Moments

Figure 6 shows the mean share gap and switching share for each of the subgroups discussed in the previous section on moments for both the data and the model. Clearly, the model does well at picking up the relative values for each of the subgroup means in both the share gap and switching share. However, it struggles to capture the absolute value, especially for the high priced NFT markets in which it underestimates the switching share and overestimates the share gap. Overall, it does a good job of picking up on the trend of blur dominance when it is the non-minting marketplace, the consistency of the share gap across subgroups, and the switching share in the cases of the parity and non-parity groups for τ and κ.

5.8 Counterfactual Analysis

To interpret the economic mechanisms embedded in the structural model and their corresponding results in alternative parameter situations, we consider a number of counterfactual situations that either alter the enforcement structure of the marketplaces or the switching frictions between them. The purpose of these counterfactual situations is not just to determine which marketplace gains market share, but to determine the settings in which enforcement and switching costs are the most important. There are five counterfactual situations we consider:

1. **OpenSea matches Blur enforcement**: We consider the situation where every market in which Blur has a higher enforcement credibility, OpenSea matches it so that $\kappa_O^{cf} = \max\{\kappa_O, \kappa_B\}$ and $\kappa_B^{cf} = \kappa_B$.

2. **Full Enforcement on Both Marketplaces**: We consider the case that Blur and OpenSea both set perfect enforcement so that $\kappa_O = \kappa_B = 1$.

3. **Remove switching frictions**:We consider the situation in which switching is frictionless so we set all switching costs to 0 so that $k_0 = 0$ and $k_1 = 0$.

4. **Blur matches OpenSea enforcement**: We consider the situation where every mar-

ket in which OpenSea has a higher enforcement credibility, Blur matches it so that $\kappa_O^{cf} = \kappa_O$ and $\kappa_B^{cf} = \max\{\kappa_O, \kappa_B\}$.

5. **Zero enforcement preference**: We consider the situation when collectors act the same as speculators in that they act only sensitive to total costs so that $\phi_K = 0$.

5.8.1 Aggregate Effects

The average effects on the share of blur, the share gap, and the switching share is presented in Table 3. In general, as can be seen in the figure, enforcement changes have little aggregate results in platform allocation of secondary market volume. Forcing OpenSea to match Blur's enforcement raises mean Blur share by 0.003. Additionally, requiring full enforcement on both platforms increases mean Blur share by 0.0009, while forcing Blur to match OpenSea's enforcement lowers mean Blur share by only 0.0023.

Table 3: Aggregate Counterfactual Effects

Scenario	Δ Mean s_B	Δ Mean Share Gap	Δ Mean Switch
OpenSea matches Blur enforcement	0.0032	0.0065	-0.0024
Full enforcement	0.0009	0.0018	-0.0023
Remove switching frictions	-0.0599	-0.1199	0.0815
Blur matches OpenSea enforcement	-0.0023	-0.0047	0.0001
Zero enforcement preference	0.0075	0.0151	-0.0034

Notes: Entries report average changes relative to the baseline model prediction.

In contrast, the switching frictions counterfactual generates much larger average effects. Removing all switching frictions reduces mean Blur share by 0.0591, reduces the share gap by 0.120, and increases the average switching share by 0.081. This contrast points to the result that off-mint switching frictions are quantitatively more important than enforcement differences in explaining the unconditional distribution of platform activity. Finally, with the collector's preferences counterfactual, eliminating the direct collector taste for enforcement increases mean Blur share by 0.0071 and reduces mean switching by 0.0033, indicating that enforcement preferences affect platform allocation, but less so than the switching frictions when averaged across the full sample.

5.8.2 Group-Specific Counterfactual Effects

Instead of simply considering the aggregate counterfactual effects, we also consider their specific effects to subgroup markets within the data. Although the average enforcement effects are small, the group-specific counterfactuals reveal that this is mostly because many markets are weakly exposed to enforcement changes. Once markets are grouped by royalty intensity, enforcement-gap direction, and mint structure, the effects become much more meaningful.

Table 4: Structured Enforcement Counterfactuals

Scenario	Market Structure	n	Δ Mean s_B
Zero enforcement preference	Medium, Blur stronger, Blur mint	819	-0.0226
Zero enforcement preference	Medium, OpenSea stronger, Blur nonmint	306	0.0509
Zero enforcement preference	High, Blur stronger, Blur mint	112	-0.0280
Zero enforcement preference	High, OpenSea stronger, Blur nonmint	64	0.0448
Full enforcement	High, Blur stronger, Blur mint	112	0.0449
Full enforcement	High, Blur stronger, Blur nonmint	15	0.0514
Full enforcement	High, OpenSea stronger, Blur mint	221	-0.0304
Full enforcement	High, OpenSea stronger, Blur nonmint	64	-0.0514

Notes: Table highlights the strongest enforcement-related counterfactual effects by market structure. The Market structure is defined in terms of 1. royalty rate groups: low ($\alpha_c < 2.5$), medium ($2.5 \leq \alpha_c \leq 7.5$), and high ($\alpha_c > 7.5$). 2. Which platform has a stronger enforcement credibility. 3. The minting platform for the collection.

The pattern observed is that enforcement credibility matters most in high-royalty markets. In the full enforcement counterfactual, high-royalty markets where Blur initially has the stronger enforcement has large increases in predicted Blur share. In contrast, when OpenSea initially has the stronger initial enforcement, removing the disadvantage for Blur has the opposite effect in the corresponding counterfactuals. These results imply that enforcement differences matter economically only when the expected royalty payment is large, so those differences translate into meaningful changes in effective cost. A second result with these counterfactuals is that the sign of the share changes to Blur depends on which platform has the initial enforcement advantage. When Blur is stronger on enforcement, forcing OpenSea to become stronger on enforcement, with the existence of a majority of speculators, results in a higher predicted share of secondary volume to trade on Blur. The opposite happens when OpenSea has the initial stronger enforcement, lining up well with the theoretical explanation earlier in the paper that enforcement is only beneficial to the platform in terms of secondary market profit if collectors value it enough.

Additionally, when it comes to enforcement preferences by collectors, the results show a clear asymmetry when these preferences are set to zero. In medium and high royalty markets, when Blur has stronger enforcement, there is a mean reduction in secondary market share when it is the minting platform. This is likely because removing preferences for enforcement shuts down a channel by which collectors choose to stay on the minting platform. In contrast, when OpenSea is the stronger enforcement platform and Blur is the non-minting marketplace, there is a positive change in the market share of Blur. This indicates that part of the likely platform advantage that OpenSea has is its ability for the retention of collectors through the enforcement channel.

Table 5: Structured Switching Counterfactuals

Scenario	Market Structure	n	Δ Mean s_B
Remove switching frictions	High price, Blur mint, non-parity	1805	-0.1448
Remove switching frictions	High price, Blur mint, tau parity	1505	-0.1538
Remove switching frictions	High price, Blur nonmint, non-parity	200	0.2041
Remove switching frictions	High price, Blur nonmint, tau parity	152	0.2191

Notes: Table highlights the strongest switching-related counterfactual The Market structure is defined in terms of 1. High-priced and low-priced markets. 2. The minting platform for the collection. 3. Whether the markets are tau-parity ($\Delta\tau \leq 0.5$) or non-parity ($\Delta\tau > 0.5$).

The switching frictions counterfactual analysis on the subgroups reveals a couple meaningful patterns. For high-price markets where Blur is the minting platform, removing switching frictions lowers Blur share by 0.14 in non-parity markets and by 0.15 in τ parity markets. By contrast, in high-price markets when Blur is not the minting platform, the same counterfactual raises Blur share by 0.20 in non-parity markets and by 0.22 in τ parity markets. These are huge effects compared to the rest of the model's counterfactual results. The results indicate that off-mint switching frictions are a major reason why market allocation in expensive NFT collections remains advantaged towards the mint platform and that this allocation is further amplified by total effective costs to traders.

6 Conclusion

This paper's focus is on studying how royalty enforcement policies and switching frictions shape platform competition, trader routing, and market outcomes in NFT markets. The theoretical section of the paper develops a strong three-period model that allows for heterogeneous trader motives, creator decisions for minting and primary price selection, and platform enforcement selection. The theoretical analysis presented in the paper shows that royalty enforcement in secondary markets is not upheld solely by the smart contracts that NFTs are built upon. Instead, equilibrium enforcement depends on the interactions in platform competition, trader-type composition, and switching frictions. In environments dominated by speculators and with little cost to switching across marketplaces, competition creates strong downward pressure on enforcement credibility. When collectors matter sufficiently or switching costs are positive, non-zero interior enforcement equilibria and asymmetric platform outcomes can be sustained.

To take these theoretical mechanisms to real NFT market data, the paper designs an empirical structural model motivated by asking how NFT trading volume in secondary markets is allocated between Blur and OpenSea as a function of royalty enforcement and switching frictions away from the mint platform. The results of the estimation suggest that there are three main takeaways. Firstly, traders are sensitive to effective total costs, so higher platform fees and stronger realized royalty burdens reduce a platform's attractiveness. Second, collectors place positive value on enforcement credibility, which means that stronger royalty enforcement can increase platform demand even when it raises direct trading costs. Third, switching costs are important and rise with price, implying that off-mint trading is especially costly in high-priced NFT markets. The model also reproduces the main aggregate relationships in the data, including the negative association between Blur's relative market share and its relative effective costs and the importance of mint structure for switching outcomes.

The counterfactual outcomes presented in the paper additionally clarify which mechanisms matter the most to both enforcement and switching frictions. At the aggregate level, changes in enforcement credibility produce somewhat small average changes in platform shares. On the other hand, removing switching frictions has much larger effects on both market allocation and off-mint trading. This indicates that switching frictions are the more meaningful pathway in explaining unconditional platform outcomes. However, the group-specific counterfactual outcomes provide more insight into where both enforcement and switching costs matter the most. Enforcement is most relevant in those in high-royalty markets and in markets where one platform holds an enforcement advantage. In these markets, changes in enforcement can meaningfully move Blur's predicted market share. The sign of the effect depends on which platform initially has the stronger enforcement regime, which is consistent with the theoretical tradeoff between collector attraction and higher effective trading costs. Additionally, removing the collector's preference for enforcement shows that part of platform advantage is transmitted through the enforcement-preference pathway, especially in mid and high-royalty markets.

Considering the results from both the theoretical and empirical components of this paper, it is clear that the economics of optional royalties depend fundamentally on the consideration of enforcement, routing, and switching. Optional and weakly enforced royalty regimes can attract secondary market volume by lowering total costs for speculative trading, but they do not fully eliminate royalties as an economic force because collectors, platform policies, and switching frictions continue to sustain some degree of enforcement and creator revenue. Thus, the paper provides an explanation as to why the NFT market does not collapse to a zero-royalty equilibrium even under competitive pressures. Instead, the paper rationalizes the market turning into a heterogeneous structure in which the value of enforcement depends on trader motives, royalty intensity, and the cost to moving across platforms.

More broadly, this paper contributes to the existing literature in the theoretical design of NFT markets and contractual structures of royalties in secondary markets by creating a testable theoretical model which combines trader routing, enforcement decisions, and multi-platform competition. It shows concretely that royalty policies in the secondary market cannot solely be understood as a creator revenue maximization problem or a platform profit problem. Rather, the paper shows that royalty enforcement in secondary markets is to be understood within a richer platform-competition model in which platforms not only compete based on their fee structures but also their ability to honor the contracts on which NFTs are built. This perspective helps explain both the race to the bottom in enforcement observed after 2022 in the NFT market and the persistence of partial compliance to secondary market royalties by platforms in the NFT market today.

There are many directions for future work based on this paper. Firstly, the choice of mint-platform by creators could be made endogenous rather than assuming that creators choose to mint on specific platform or either platform. Second, a richer structure for the heterogeneity may be introduced by further building upon the existing literature of informed and uninformed traders beyond speculators and collectors, where informational and quality signals, dependent on the collection-level attributes, are considered within the routing and enforcement framework. Lastly, more work could be done to consider the implications of enforcement regimes and multiple NFT market structures on creator welfare, a direction motivated by the observered drop in the growth of minting NFTs by creators. This includes consideration to the long-run consequences of creator participation with the choice of a specific type of model of platform royalty compliance.

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A Proofs of Propositions

A.1 Proof of Proposition 1 (Monopoly Case)

Proof. For a monopoly platform M with a mandatory royalty regime, let total trading activity on the platform be summarized by the endogenous primary market quantity $Q_I(\kappa)$ and price p_I as well as the endogenously determined secondary market quantity $Q_{II}(\kappa)$ and price $p_{II}(\kappa)$. Consider the single platform choosing enforcement credibility $\kappa \in [0, 1]$.

Platform profit is given by:

$$\Pi(\kappa) = \beta \left[p_I Q_I(\kappa) + \delta_M p_{II}(\kappa) Q_{II}(\kappa) \right] - C(\kappa).$$

Assume $Q_I(\kappa), p_{II}(\kappa), Q_{II}(\kappa)$ are differentiable in κ. Differentiating with respect to κ, we find:

$$\Pi'(\kappa) = \beta \left[p_I Q_I'(\kappa) + \delta_M \left(p_{II}'(\kappa) Q_{II}(\kappa) + p_{II}(\kappa) Q_{II}'(\kappa) \right) \right] - C'(\kappa).$$

An interior optimum solution, $\kappa^* \in (0,1)$ must satisfy $\Pi'_M(\kappa^*) = 0$. That is, it must be the cast that

$$\beta\left[p_I Q'_I(\kappa) + \delta_M\left(p'_{II}(\kappa)Q_{II}(\kappa) + p_{II}(\kappa)Q'_{II}(\kappa)\right)\right] = C'(\kappa)$$

Thus, this condition states that the marketplace platform will increase enforcement credibility of royalties κ until the marginal fee revenue from the primary and secondary market is equal to the marginal cost of enforcement. Now, consider the first order condition at the boundary at $\kappa = 0$:

$$\Pi'_M(0) = \frac{d}{d\kappa}(\beta\left[p_I Q_I(\kappa) + \delta_M p_{II}(\kappa)Q_{II}(\kappa)\right])|_{\kappa=0} - C'(0)$$

If $\Pi'_M(0) > 0$, then profit is increasing when enforcement credibility is chosen to be $\kappa = 0$. Since Π_M is differentiable at 0, there exists $\varepsilon > 0$ such that

$$\Pi_M(\varepsilon) > \Pi_M(0)$$

Thus, $\kappa = 0$ cannot be an optimal choice. Because Π_M is continuous and the feasible set $[0,1]$ is compact, the Extreme Value Theorem implies that there exists some optimal $\kappa^* \in [0,1]$ such that

$$\Pi_M(\kappa^*) \geq \Pi_M(\kappa) \quad \text{for all } \kappa \in [0,1].$$

Since $\kappa = 0$ is not optimal, it follows that $\kappa^* > 0$. If additionally, platform profit is strictly concave in κ, that is $\Pi''_M(\kappa) < 0$ for all $\kappa \in (0,1)$, then this optimum is unique.

If instead, $\Pi'_M(0) \leq 0$, then a marginal increase in enforcement of royalties from zero does not improve profit. Under concavity of the profit of the platform, profit is then maximized at the boundary $k^* = 0$.

$\square$

A.2 Proof of Proposition 2 (Race to the Bottom)

Proof. Consider a symmetric duopoly with two platforms $m \in \{A, B\}$ which are both under a mandatory royalty regime with a fixed royalty rate $\alpha > 0$, identical platform fees β, and no switching costs between the platforms. Additionally, the marketplace for NFTs is solely driven by speculative trading behavior.

Under the mandatory royalty regime, the effective total trading cost on platform m is given by

$$\tau_m = \beta + \kappa_m \alpha$$

By assumption, β and α are identical across platforms. Thus, any cost differences between the two platforms is driven solely be the marketplaces enforcement credibility κ_m. Speculators purely cost minimizing, and thus, they route all of their trading volume to whichever platform gives them the lowest total trading cost. In the limiting case of the exponential routing model introduced in the general model (that with $\lambda \to \infty$), the platform with the lowest total cost captures all the trading volume while platforms with higher costs get none. If costs are equal, the platforms will get an equivalent market share that is half of the total volume. Thus,

$$s_A = \begin{cases} 1 & \text{if } \tau_A < \tau_B, \\ 1/2 & \text{if } \tau_A = \tau_B, \\ 0 & \text{if } \tau_A > \tau_B, \end{cases} \qquad s_B = 1 - s_A$$

Let $\kappa_{\min} = \min\{\kappa_A, \kappa_B\}$. Since all trade occurs on the lowest-cost platform, equilibrium quantities in the primary and secondary markets, including primary demand Q_I, secondary price p_{II}, and total volume Q_{II}, depend only on $\tau_{\min} = \beta + \kappa_{\min}\alpha$. The higher-enforcement platform receives zero secondary trading volume.

Platform m's profit in the secondary market is given by

$$\Pi_m = \beta Q_{II,m} - C_m(\kappa_m),$$

where $C_m(\kappa_m)$ is the enforcement cost with $C_m(\kappa_m) > 0$ for $\kappa_m > 0$.

If $\kappa_m > \kappa_{-m}$, then platform m receives no trading volume. If $\kappa_m < \kappa_{-m}$, platform m captures all trading volume and earns positive fee revenue.

Consider the symmetric equilibrium with $\kappa_A = \kappa_B = \kappa > 0$. Platform A can deviate to $\kappa'_A = \kappa - \varepsilon$ for arbitrarily small $\varepsilon > 0$. This downward pressure on κ_A lowers enforcement costs and thus, lowers its trading cost, so $\tau'_A < \tau_B$, which means that platform A captures all trading volume. Since enforcement costs are lower and revenue increases, this strategy is profitable. Therefore, no symmetric equilibrium with $\kappa > 0$ can exist.

Finally, consider $\kappa_A = \kappa_B = 0$. In this case, both platforms have identical costs and split the total volume in the market, so each platform receives half of the total trading volume. Any deviation by a platform m to $\kappa_m > 0$ increases the platform's total cost, causing it to lose all speculative traders. Since $C_m(\kappa_m) > 0$, this strategy must have strictly negative profit. Therefore, no profitable deviation exists from $(\kappa_A, \kappa_B) = (0, 0)$. Thus, $(\kappa^*_A, \kappa^*_B) = (0, 0)$ is the unique Nash equilibrium.

□

A.2.1 Closed-form Example with Uniform Valuations

Secondary buyers have valuations $v_{II} \sim U[0, 1]$, generating demand

$$Q_{II}(p) = 1 - p.$$

Assume each token is resold once, so market clearing gives

$$Q_{II} = Q_I = 1 - p_{II} \qquad \Rightarrow \qquad p_{II} = 1 - Q_I.$$

Speculator profit is based only on resale, not consumption:

$$\pi_i = (1 - \tau_{\min})p_{II} - p_I - c_i = (1 - \tau_{\min})(1 - Q_I) - p_I - c_i.$$

The entry condition is

$$\pi_i \geq 0 \iff c_i \leq (1 - \tau_{\min})(1 - Q_I) - p_I.$$

Since $c_i \sim U[0, 1]$, interior primary demand is

$$Q_I = (1 - \tau_{\min})(1 - Q_I) - p_I.$$

Solving the fixed point gives

$$Q_I(\tau_{\min}, p_I) = \frac{(1 - \tau_{\min}) - p_I}{2 - \tau_{\min}}.$$

Hence the secondary price is

$$p_{II}(\tau_{\min}, p_I) = 1 - Q_I = \frac{1 + p_I}{2 - \tau_{\min}}.$$

The creator's revenue comes from net primary-market sales and discounted royalties:

$$\Pi_C = (1 - \beta)p_I Q_I + \delta_C \alpha \kappa_{\min}\, p_{II} Q_I.$$

Define

$$X = 1 - \tau_{\min}, \qquad Y = 2 - \tau_{\min}, \qquad Z = \alpha\kappa_{\min},$$

so that

$$Q_I = \frac{X - p_I}{Y}, \qquad p_{II} = \frac{1 + p_I}{Y}.$$

Then creator profit is

$$\Pi_C(p_I, \kappa_{\min}) = \frac{1}{Y}(X - p_I)\left[(1-\beta)p_I + \frac{\delta_C Z(1+p_I)}{Y}\right].$$

Let

$$K = (1-\beta) + \frac{\delta_C Z}{Y}, \qquad L = \frac{\delta_C Z}{Y}.$$

Then

$$\Pi_C(p_I) = \frac{1}{Y}(X - p_I)(Kp_I + L).$$

The first-order condition gives the optimal primary price

$$p_I^*(\kappa_{\min}) = \frac{KX - L}{2K}.$$

The quantities are then

$$Q_I^*(\kappa_{\min}) = \frac{X - p_I^*(\kappa_{\min})}{Y}, \qquad p_{II}^*(\kappa_{\min}) = \frac{1 + p_I^*(\kappa_{\min})}{Y}.$$

Since each token is resold once,

$$Q_{II}^* = Q_I^*(\kappa_{\min}).$$

Because all volume is routed to the lower-enforcement platform, platform A's profit is

$$\Pi_A(\kappa_A, \kappa_B) = \begin{cases} \beta\Big(p_I^* Q_I^* + \delta_M p_{II}^* Q_{II}^*\Big) - C(\kappa_A), & \kappa_A < \kappa_B, \tfrac{1}{2}\beta\Big(p_I^* Q_I^* + \delta_M p_{II}^* Q_{II}^*\Big) - C(\kappa_A), \\ \kappa_A = \kappa_B, -C(\kappa_A), & \kappa_A > \kappa_B. \end{cases}$$

Since a lower κ decreases $\tau_{\min}$, it raises equilibrium trade volume and enforcement cost $C(\kappa)$ is increasing in κ. Therefore, each platform has an incentive to undercut any positive enforcement level. Therefore, the best responses are

$$BR_A(\kappa_B) = 0, \qquad BR_B(\kappa_A) = 0.$$

Thus the unique Nash equilibrium is

$$\kappa_A^* = \kappa_B^* = 0.$$

A.3 Proof of Proposition 3 (Collectors)

Proof. Consider a duopoly with marketplaces $m \in A, B$. Each platform chooses enforcement credibility $\kappa_m \in [0, 1]$. There is a mandatory royalty regime, the royalty rate $\alpha > 0$ is fixed, and platform fees are symmetric at β. The market consists of both speculators and collectors as traders and routing between marketplaces follows the logit structure in the general model.

Let platform m's profit be given by

$$\Pi_m(\kappa_m, \kappa_{-m}) = \beta(p_I Q_{I,m}(\kappa_m, \kappa_{-m}) + \delta_M p_{II} Q_{II,m}(\kappa_m, \kappa_{-m})) - C(\kappa_m)$$

where $C'(\kappa) > 0$, $C''(\kappa) > 0$ and assume that $Q_{I,m}$, $Q_{II,m}$, p_I, and p_{II} are continuously differentiable. Total trading cost is given as $\tau_m = \beta + \alpha\kappa_m$. For clarity, split the quantity demanded in primary and secondary markets into speculative and collector components.

Then each period $i \in \{I, II\}$ gives

$$Q_{i,m} = Q^S_{i,m}(\kappa_m, \kappa_{-m}) + Q^K_{i,m}(\kappa_m, \kappa_{-m})$$

resulting in platform profit as

$$\Pi_m(\kappa_m, \kappa_{-m}) = \beta(p_I(Q^S_{I,m}(\kappa_m, \kappa_{-m}) + Q^K_{I,m}(\kappa_m, \kappa_{-m}))$$

$$+\delta_M p_{II}(Q^S_{II,m}(\kappa_m, \kappa_{-m}) + Q^K_{II,m}(\kappa_m, \kappa_{-m})) - C(\kappa_m)$$

Differentiating with respect to a marketplaces own enforcement gives

$$\frac{\partial \Pi_m}{\partial \kappa_m} = \beta\Bigg[\frac{\partial p_I}{\partial \kappa_m}(Q^S_{I,m} + Q^K_{I,m}) + p_I\left(\frac{\partial Q^S_{I,m}}{\partial \kappa_m} + \frac{\partial Q^K_{I,m}}{\partial \kappa_m}\right)$$

$$+\delta_M\left(\frac{\partial p_{II}}{\partial \kappa_m}(Q^S_{II,m} + Q^K_{II,m}) + p_{II}\left(\frac{\partial Q^S_{II,m}}{\partial \kappa_m} + \frac{\partial Q^K_{II,m}}{\partial \kappa_m}\right)\right)\Bigg] - C'(\kappa_m).$$

This derivative decomposes into four marginal effects:

$$\frac{\partial \Pi_m}{\partial \kappa_m} = \underbrace{\beta\Big(p_I\frac{\partial Q^S_{I,m}}{\partial \kappa_m} + \delta_M p_{II}\frac{\partial Q^S_{II,m}}{\partial \kappa_m}\Big)}_{\text{speculator effect}} + \underbrace{\beta\Big(p_I\frac{\partial Q^K_{I,m}}{\partial \kappa_m} + \delta_M p_{II}\frac{\partial Q^K_{II,m}}{\partial \kappa_m}\Big)}_{\text{collector effect}}$$

$$+ \quad \underbrace{\beta\frac{\partial p_I}{\partial \kappa_m}(Q^S_{I,m} + Q^K_{I,m})}_{\text{creator-side primary-market effect}} \quad + \underbrace{\beta\delta_M\frac{\partial p_{II}}{\partial \kappa_m}(Q^S_{II,m} + Q^K_{II,m})}_{\text{secondary-price effect}} - \quad \underbrace{C'(\kappa_m)}_{\text{direct enforcement-cost effect}} \quad .$$

By construction, the speculator-loss effect is negative so:

$$\frac{\partial Q^S_{I,m}}{\partial \kappa_m} < 0, \qquad \frac{\partial Q^S_{II,m}}{\partial \kappa_m} < 0.$$

Similarly, the collector-retention effect is positive so:

$$\frac{\partial Q^K_{I,m}}{\partial \kappa_m} > 0, \qquad \frac{\partial Q^K_{II,m}}{\partial \kappa_m} > 0.$$

The creator chooses p_I and minting decisions in period 0 by anticipating future royalty realization in period 2. Stronger enforcement raises the expected fraction of future royalties that the creator should receive. Following the logic of Zou, Gu, Liu (2024), it is clear that stronger royalty expectation tends to reduce the primary market price chosen by the creator. Thus, the creators primary-market effect may be negative to firm profit if enforcement goes up. However, a lower mint price may increase participation in the primary market, thus the third marginal effect is generally ambiguous without concrete structure.

Similarly, the sign of the secondary market price effect is also ambiguous as stronger enforcement can raise secondary prices by improving the collector interest or can lower prices by discouraging speculative demand.

Now consider the possibilities of a symmetric result for enforcement credibility (κ, κ). Then we can define the function:

$$\hat{\Pi}(\kappa) := \Pi_m(\kappa, \kappa)$$

By assumption, we know that $\hat{\Pi}$ is continuously differentiable on $\kappa \in [0, 1]$. Now, consider the first-order condition for this function $\hat{\Pi}'$. Clearly, if $\left.\frac{\partial \Pi_m}{\partial \kappa_m}\right|_{(0,0)} > 0$, then $\hat{\Pi}'(0) > 0$ so by differentiability there exists some $\epsilon > 0$ such that $\hat{\Pi}(\epsilon) > \hat{\Pi}(0)$, so $\kappa = 0$ cannot mazimize symmetric platform profit. Also, if $\left.\frac{\partial \Pi_m}{\partial \kappa_m}\right|_{(1,1)} < 0$, then by a similar argument, we know that the optimal symmetric pair cannot be $\kappa = 1$. Thus, because $\hat{\Pi}'$ is continuous and changes sign, the Intermediate Value Theorem implies that there exists some $\kappa^* \in (0, 1)$ such that $\hat{\Pi}'(\kappa^*) = 0$ so that $\kappa_A = \kappa_B = \kappa^*$ is a symmetric interior equilibrium.

$\square$

A.3.1 Example Closed Form Existence of a Symmetric Equilibrium

Assume the following:

- Secondary buyers have valuations $v_{II} \sim U[0,1]$
- Speculator participation costs are $c_i \sim U[0,1]$
- Collectors value enforcement directly through $\phi > 0$
- Two marketplaces A, B have common platform fee β
- Royalty rate is fixed at α
- Enforcement cost function is given as $C(\kappa) = \frac{c}{2}\kappa^2$ with $c > 0$
- A fraction θ of the traders are speculators and $(1-\theta)$ are collectors
- Collectors buy during the primary market and hold for consumption in the secondary market

Consider the case of a possible symmetric equilibrium. In this case, each platform receives half of the total fee generating volume and gets the same amount of each time of buyer. Under mandatory royalties, total trading cost is $\tau = \beta + \alpha\kappa$. Speculators buy in Period 1 only for resale in Period 2 while collectors buy for ownership and do not resell. Secondary-market supply is given by speculator quantity:

$$Q_{II} = Q_I^S.$$

Market clearing therefore implies

$$Q_I^S = 1 - p_{II} \qquad \Rightarrow \qquad p_{II} = 1 - Q_I^S.$$

A speculator's expected payoff from buying in Period 1 is the net resale payoff in Period 2:

$$\pi_i^S = (1-\tau)p_{II} - p_I - c_i.$$

We can define

$$X(\kappa) := 1 - \tau = 1 - \beta - \alpha\kappa.$$

Using $p_{II} = 1 - Q_I^S$, this becomes

$$\pi_i^S = X(\kappa)(1 - Q_I^S) - p_I - c_i.$$

The speculator enters if and only if $\pi_i^S \geq 0$, so

$$c_i \leq X(\kappa)(1 - Q_I^S) - p_I.$$

Since $c_i \sim U[0,1]$, interior speculator participation is

$$Q_I^S = \theta\big(X(\kappa)(1 - Q_I^S) - p_I\big).$$

Solving for Q_I^S gives

$$Q_I^S(\kappa, p_I) = \frac{\theta\big(X(\kappa) - p_I\big)}{1 + \theta X(\kappa)}.$$

Define

$$Y(\kappa) := 1 + \theta X(\kappa) = 1 + \theta(1 - \beta - \alpha\kappa).$$

Then

$$Q_I^S(\kappa, p_I) = \frac{\theta(X(\kappa) - p_I)}{Y(\kappa)}.$$

Using market clearing, the secondary-market price is

$$p_{II}(\kappa, p_I) = 1 - Q_I^S(\kappa, p_I) = \frac{1 + \theta p_I}{Y(\kappa)}.$$

Period 1 utility for collectors is

$$u_i^K = v_i + \phi\kappa - p_I.$$

Since $v_i \sim U[0, 1]$, collector participation is

$$Q_I^K(\kappa, p_I) = (1 - \theta)(1 - p_I + \phi\kappa).$$

Total primary market demand is then

$$Q_I(\kappa, p_I) = Q_I^S(\kappa, p_I) + Q_I^K(\kappa, p_I).$$

Thus

$$Q_I(\kappa, p_I) = \frac{\theta(X(\kappa) - p_I)}{Y(\kappa)} + (1 - \theta)(1 - p_I + \phi\kappa).$$

Then when creator chooses the mint price p_I in Period 0 considering secondary and primary market payments. Creator profit is

$$\Pi_C(p_I, \kappa) = (1 - \beta)p_I Q_I(\kappa, p_I) + \delta_C \alpha\kappa\, p_{II}(\kappa, p_I) Q_I^S(\kappa, p_I).$$

Substituting gives

$$\Pi_C(p_I, \kappa) = (1-\beta)p_I \left[\frac{\theta(X(\kappa) - p_I)}{Y(\kappa)} + (1 - \theta)(1 - p_I + \phi\kappa)\right] + \delta_C \alpha\kappa \left(\frac{1 + \theta p_I}{Y(\kappa)}\right)\left(\frac{\theta(X(\kappa) - p_I)}{Y(\kappa)}\right).$$

Rewrite this as

$$\Pi_C(p_I, \kappa) = x(\kappa)p_I^2 + y(\kappa)p_I + z(\kappa),$$

where

$$x(\kappa) = -(1-\beta)\left(\frac{\theta}{Y(\kappa)} + 1 - \theta\right) - \frac{\delta_C \alpha\kappa\,\theta^2}{Y(\kappa)^2},$$

$$y(\kappa) = (1-\beta)\left(\frac{\theta X(\kappa)}{Y(\kappa)} + (1-\theta)(1+\phi\kappa)\right) + \frac{\delta_C \alpha\kappa\,\theta(\theta X(\kappa) - 1)}{Y(\kappa)^2},$$

$$z(\kappa) = \frac{\delta_C \alpha\kappa\,\theta X(\kappa)}{Y(\kappa)^2}.$$

Since $x(\kappa) < 0$, creator profit is concave in p_I, so the first-order condition gives the primary market price

$$p_I^*(\kappa) = -\frac{y(\kappa)}{2x(\kappa)}.$$

Substituting gives

$$Q_I^{S*}(\kappa) = \frac{\theta(X(\kappa) - p_I^*(\kappa))}{Y(\kappa)},$$

$$Q_I^{K*}(\kappa) = (1-\theta)(1 - p_I^*(\kappa) + \phi\kappa),$$

$$Q_I^*(\kappa) = Q_I^{S*}(\kappa) + Q_I^{K*}(\kappa), \qquad Q_{II}^*(\kappa) = Q_I^{S*}(\kappa),$$

and

$$p_{II}^*(\kappa) = \frac{1 + \theta p_I^*(\kappa)}{Y(\kappa)}.$$

Each platform receives one-half of the total volume. Therefore, each platform profit is

$$\hat{\Pi}(\kappa) = \frac{\beta}{2}\Big(p_I^*(\kappa)Q_I^*(\kappa) + \delta_M p_{II}^*(\kappa)Q_{II}^*(\kappa)\Big) - \frac{c}{2}\kappa^2.$$

Now choose

$$\theta = 0.3, \qquad \beta = 0.05, \qquad \alpha = 0.05, \qquad \phi = 0.5, \qquad \delta_C = \delta_M = 1, \qquad c = 0.01.$$

Under this parameterization,

$$p_I^*(0) \approx 0.49374739, \qquad p_I^*(1) \approx 0.66839286,$$

$$Q_I^{S*}(0) \approx 0.10651812, \qquad Q_I^{S*}(1) \approx 0.05471035,$$

$$Q_I^{K*}(0) \approx 0.35437682, \qquad Q_I^{K*}(1) \approx 0.58212500,$$

so that

$$Q_I^*(0) \approx 0.46089494, \qquad Q_I^*(1) \approx 0.63683535,$$

$$p_{II}^*(0) \approx 0.89348188, \qquad p_{II}^*(1) \approx 0.94528965.$$

The symmetric duopoly profit then has

$$\hat{\Pi}'(0) \approx 0.00317801 > 0, \qquad \hat{\Pi}'(1) \approx -0.00545011 < 0.$$

Because $\hat{\Pi}(\kappa)$ is continuously differentiable on $[0,1]$, the Intermediate Value Theorem implies that there exists some

$$\kappa^* \in (0,1)$$

such that

$$\hat{\Pi}'(\kappa^*) = 0.$$

Therefore, under this concrete closed-form parameterization, there exists a positive symmetric interior enforcement equilibrium

$$\kappa_A^* = \kappa_B^* = \kappa^* \in (0,1).$$

A.4 Proof of Proposition 4 (Switching Costs)

Proof. Let there be a symmetric duopoly with platforms $m \in \{A, B\}$ with a mandatory royalty regime with a fixed royalty rate $\alpha > 0$ and symmetric platform fee $\beta > 0$. There are both speculators and collectors present, and the NFT is minted by the creator on the platform d with the non-minting platform being $n \neq d$. Assume that positive switching costs enter the routing problem through an exogenous switching-cost parameter $k > 0$ paid when a trader routes a secondary-market trade away from the minting platform..

By definition, market share of each trader type t in the secondary market is defined as

$$s_{m,t} = \frac{\exp(\lambda_t \bar{u}_t(m))}{\sum_{l \in \mathcal{M}} \exp(\lambda_t \bar{u}_t(l))}$$

Consider speculators, we can define the utility for each platform for the speculators as

$$u_{i,d}^S = (1 - \tau_d)p_{II} - p_I - c_i$$

$$u_{i,n}^S = (1 - \tau_n)p_{II} - p_I - kp_{II} - c_i$$

Thus, we define

$$u_{i,d}^S - u_{i,n}^S = p_{II}(\alpha(\kappa_n - \kappa_d) + k)$$

.

Then

$$s_{d,S} = \frac{\exp(\lambda_s u_{i,d}^S)}{\exp(\lambda_s u_{i,d}^S) + \exp(\lambda_s u_{i,n}^S)} = \frac{\exp(\lambda_s u_{i,d}^S)}{\exp(\lambda_s u_{i,d}^S)(1 + \frac{\exp(\lambda_s u_{i,n}^S))}{\exp(\lambda_s u_{i,d}^S)})}$$

$$= \frac{1}{1 + \exp[\lambda_S(u_{i,n}^S - u_{i,d}^S)]} = \frac{1}{1 + \exp[(-\lambda_S(u_{i,d}^S - u_{i,n}^S))]}$$

so the resulting market share for the home platform is

$$s_{d,S} = \frac{1}{1 + \exp[-\lambda_S(p_{II}(\alpha(\kappa_n - \kappa_d) + k))]}$$

Similarly for collectors, we have

$$u^K_{i,d} = v_i + \phi_K \kappa_d - p_I - p_{II}(\tau_d)$$

$$u^K_{i,n} = v_i + \phi_K \kappa_n - p_I - p_{II}(\tau_n) - kp_{II}$$

and

$$u^K_{i,d} - u^K_{i,n} = (\phi_K - \alpha p_{II})(\kappa_d - \kappa_n) + kp_{II}$$

Thus, for collectors the market share of the home platform is

$$s_{d,K} = \frac{1}{1 + \exp[-\lambda_K((\phi_K - \alpha p_{II})(\kappa_d - \kappa_n) + kp_{II})]}$$

Now consider the case when the platforms choose symmetric enforcement so that $\kappa_n = \kappa_d = \kappa$. Then the market shares reduce to (holding the secondary-market price fixed)

$$s_{d,S} = \frac{1}{1 + \exp[-\lambda_S kp_{II}]} \qquad s_{d,K} = \frac{1}{1 + \exp[-\lambda_K kp_{II}]}$$

By assumption, we know that $\lambda_S \geq \lambda_K > 0$. Thus, if $k > 0$, which means that there is positive switching costs, then the result is

$$s_{d,S} > \frac{1}{2} \qquad s_{d,K} > \frac{1}{2}$$

so there is minting platform bias for selection of trading in the secondary market by both the collectors and speculators.

Now, consider the change in enforcement's effect on the routing share of collectors and speculators. For speculators, at a symmetric interior enforcement credibility level for the duopoly, we have that:

$$\frac{\partial s_{d,S}}{\partial \kappa_d} = s_{d,S}(1 - s_{d,S})\frac{\partial}{\partial \kappa_d}(\lambda_S(p_{II}(\alpha(\kappa_n - \kappa_d) + k))) = -\lambda_S \alpha p_{II} s_{d,S}(1 - s_{d,S})$$

Because the routing decision comes after a secondary market price, we consider it fixed. Since, $p_{II} > 0$, $0 < s_{d,S} < 1$, and $\lambda_S > 0$, we have that $\frac{\partial s_{d,S}}{\partial \kappa_d} < 0$ or simply that the share of speculators in the minting platform is strictly decreasing with an increase to enforcement credibility. For collectors, we get

$$\frac{\partial s_{d,K}}{\partial \kappa_d} = s_{d,K}(1 - s_{d,K})\frac{\partial}{\partial \kappa_d}(\lambda_K((\phi_K - \alpha p_{II})(\kappa_d - \kappa_n) + k p_{II})) = \lambda_K(\phi_K - \alpha p_{II}) s_{d,K}(1 - s_{d,K})$$

Since we know that $\lambda_K > 0$ and $0 < s_{d,K} < 1$, then we know that if $\phi_K > \alpha p_{II}$ (which simply means that collectors value enforcement more than the royalty payment), then we have $\frac{\partial s_{d,K}}{\partial \kappa_d} > 0$. Thus, in such a case the primary minting platform's secondary market share of collectors is increasing in enforcement.

Now consider how the routing response changes as the exogenous switching-cost parameter k varies in addition to changing their enforcement credibility. We have for speculators that

$$\frac{\partial^2 s_{d,S}}{\partial \kappa_d \partial k} = \frac{\partial}{\partial \kappa_d}(-\lambda_S \alpha p_{II} s_{d,S}(1 - s_{d,S})) = -\lambda_S \alpha p_{II} \frac{\partial}{\partial k}(s_{d,S}(1 - s_{d,S})) = -\lambda_S \alpha p_{II}(1 - 2s_{d,S})\frac{\partial s_{d,S}}{\partial k}$$

$$= -\lambda_S \alpha p_{II}(1 - 2s_{d,S})(\lambda_S p_{II}) s_{d,S}(1 - s_{d,S}) = (\lambda_S p_{II})^2 \alpha s_{d,S}(1 - s_{d,S})(2s_{d,S} - 1)$$

and since we showed that at a symmetric enforcement level, we have $s_{d,S} > \frac{1}{2}$, then we know that $\frac{\partial^2 s_{d,S}}{\partial \kappa_d \partial k} > 0$. Therefore, at a symmetric enforcement level, stronger or increasing switching costs reduce the sensitivity of speculative routing to enforcement differences between platforms. Similarly, for collectors we have

$$\frac{\partial^2 s_{d,K}}{\partial \kappa_d \, \partial k} = -\lambda_K^2 p_{II}(\phi_K - \alpha p_{II}) \, s_{d,K}(1 - s_{d,K})(2s_{d,K} - 1)$$

.

Thus, if collectors value enforcement less than the royalty payment, we have $\frac{\partial^2 s_{d,K}}{\partial \kappa_d \, \partial k} > 0$. Additionally, if we consider the platform profit for the minting platform, given by

$$\Pi_d(\kappa_d, \kappa_n, k) = \beta(p_I Q_{I,d} + \delta_M p_{II} Q_{II,d}) - C(\kappa_d)$$

.

Then differentiating with respect to enforcement credibility gives

$$\frac{\partial \Pi_d}{\partial \kappa_d} = \beta \left[\frac{\partial p_I}{\partial \kappa_d} Q_{I,d} + p_I \frac{\partial Q_{I,d}}{\partial \kappa_d} + \delta_M \left(\frac{\partial p_{II}}{\partial \kappa_d} Q_{II,d} + p_{II} \frac{\partial Q_{II,d}}{\partial \kappa_d} \right) \right] - C'(\kappa_d).$$

Additionally, we know that, assuming that the share of speculators in the market is θ, that

$$Q_{II,d} = \theta s_{d,S} Q_{II,S} + (1 - \theta) s_{d,K} Q_{II,K}$$

so

$$\frac{\partial Q_{II,d}}{\partial \kappa_d} = \theta Q_{II,S} \frac{\partial s_{d,S}}{\partial \kappa_d} + (1 - \theta) Q_{II,K} \frac{\partial s_{d,K}}{\partial \kappa_d}$$

From the routing results established above,

$$\frac{\partial s_{d,S}}{\partial \kappa_d} < 0 \qquad \text{and} \qquad \frac{\partial^2 s_{d,S}}{\partial \kappa_d \, \partial k} > 0.$$

Thus, stronger enforcement by the mint platform reduces the speculative routing share, but higher switching costs make this negative effect smaller. Therefore, as k rises, the speculative-routing penalty from stronger enforcement is weakened. Returning to the marginal return to platform profitability in own enforcement, we can determine how switching costs affect this marginal payoff, differentiate once more with respect to k. Define

$$\Pi_{\kappa k} = \frac{\partial^2 \Pi_d}{\partial \kappa_d \, \partial k}.$$

The interpretation of $\Pi_{\kappa k}$ is to determine whether increasing switching costs increases or decreases the return to enforcement of the minting platform. Since $\frac{\partial^2 s_{d,S}}{\partial \kappa_d \partial k} > 0$, then a higher k makes $\frac{\partial s_{d,S}}{\partial \kappa_d}$ less negative so it makes $\frac{\partial Q_{II,d}}{\partial \kappa_d}$ less negative which means $\frac{\partial Q_{II,d}}{\partial \kappa_d \partial k} > 0$ which results in $\Pi_{\kappa k} > 0$ when the contribution to marginal changes in enforcement from switching costs outweigh the marginal effects through prices, primary market quantities and collector secondary market volume (or when collectors value enforcement less than secondary market royalty payments).

Now, suppose that there exists a symmetric interior enforcement equilibrium $\kappa_d = \kappa_n = \kappa^*(k)$ so that both platforms choose the same enforcement level which depends on switching costs. The first-order condition for a symmetric equilibrium to exist is $\frac{\partial \Pi_d}{\partial \kappa_d}(\kappa^*(k), \kappa^*(k), k) = 0$. Then we can define the function

$$F(\kappa, k) = \frac{\partial \Pi_d}{\partial \kappa_d}(\kappa, \kappa, k)$$

which means that we can write the first-order condition as $F(\kappa^*(k), k) = 0$ Then differentiating this result with respect to k gives

$$\frac{d}{dk}F(\kappa^*(k), k) = F_\kappa(\kappa^*(k), k)\frac{d\kappa^*(k)}{dk} + F_k(\kappa^*(k), k) = 0.$$

This is the same as writing

$$\frac{d\kappa^*(k)}{dk} = \frac{-F_k(\kappa^*(k), k)}{F_\kappa(\kappa^*(k), k)} = \frac{-\Pi_{\kappa k}}{\Pi_{\kappa\kappa}}$$

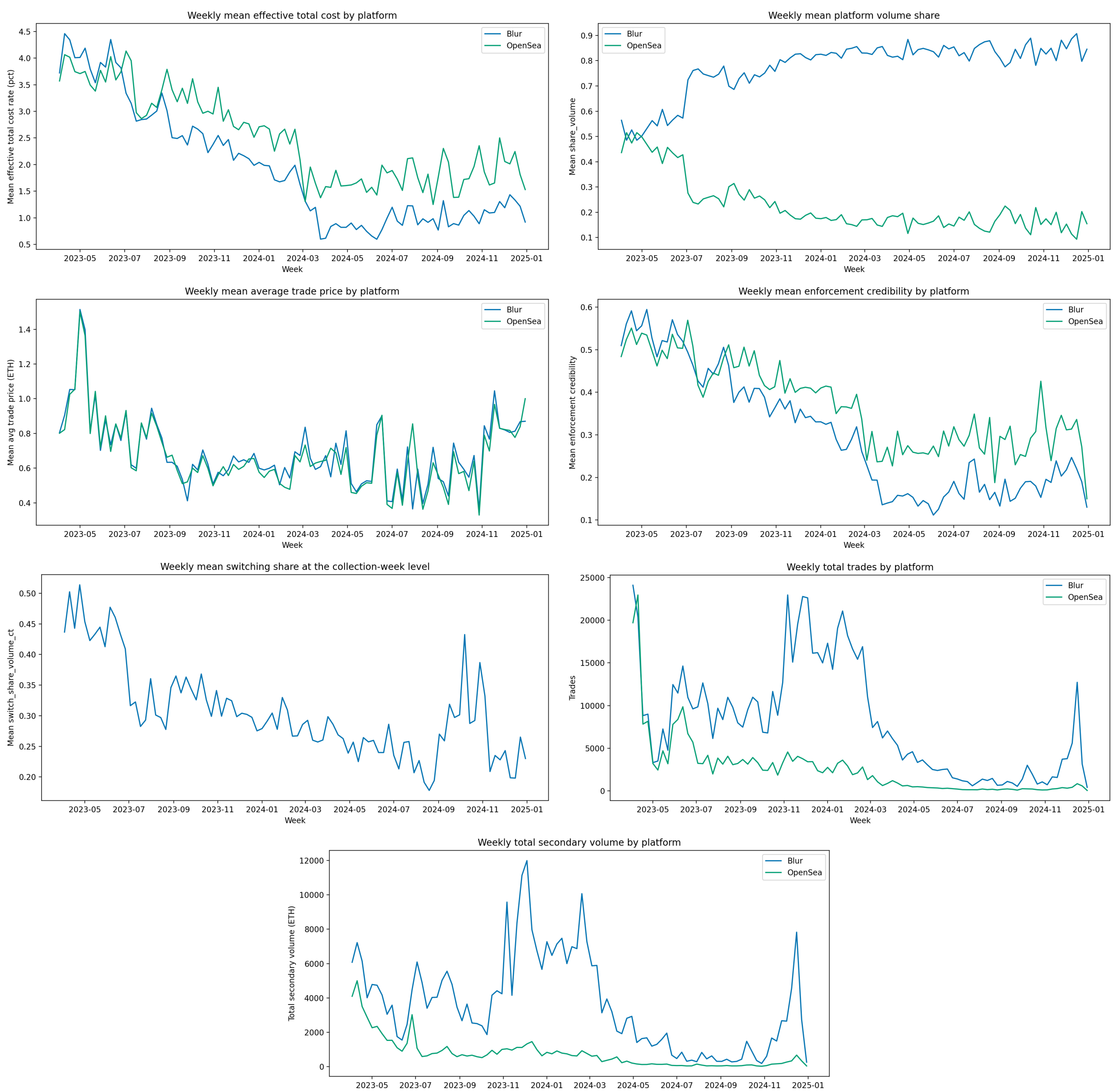
and since we know that, given the marginal changes in enforcement from switching costs outweigh the marginal effects through prices or primary market quantities, we have $\Pi_{\kappa k} > 0$ so $F_k(\kappa^*(k), k) > 0$. Additionally, if we assume that platforms have strict concavity in their own enforcement, which means that there is diminishing marginal returns to increasing enforcement from the minting platform, then $\Pi_{\kappa\kappa} < 0$ so $F_\kappa(\kappa^*(k), k) < 0$. Then $\frac{d\kappa^*(k)}{dk} > 0$. Therefore, the symmetric equilibrium enforcement level is increasing in switching costs.

□

B Empirical Results

B.1 Summary of NFT Data

Figure 7: Summary Plots of Post-Cleaning Data on Weekly Basis



B.2 Additional Results from Estimation

Figure 8: Scatter Plots of Fit After Estimation

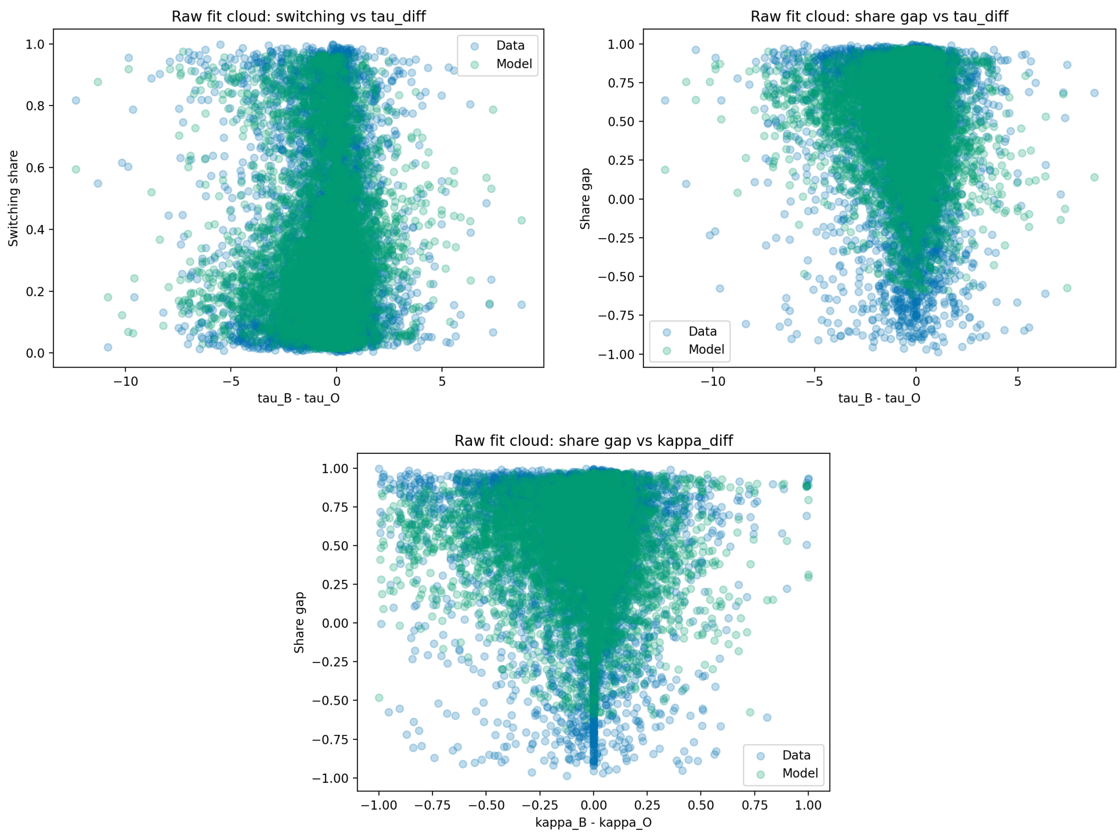
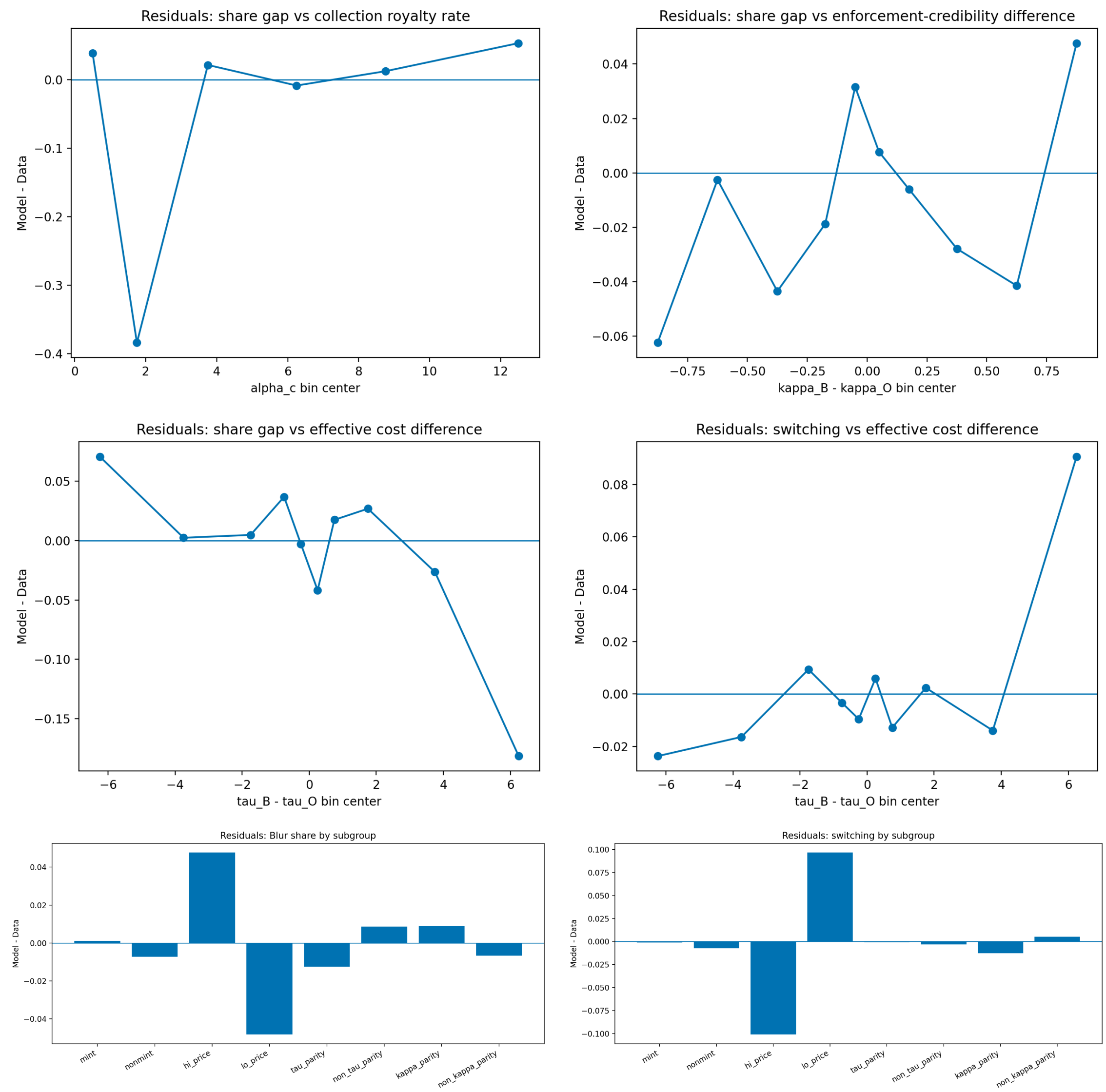


Figure 9: Residuals after Estimation



B.3 Additional Results from Counterfactual Analysis

Table 6: Sub-Group Switching Counterfactuals by Price Group, Mint Structure, and Total Cost Difference

Scenario	Price Group	Market Structure	n	Δ Mean s_B	Δ Mean Share Gap
Remove switching frictions	High	Blur mint, non-parity	1805	-0.1425	-0.2850
Remove switching frictions	High	Blur mint, tau parity	1505	-0.1515	-0.3030
Remove switching frictions	High	Blur nonmint, non-parity	200	0.1989	0.3978
Remove switching frictions	High	Blur nonmint, tau parity	152	0.2134	0.4267
Remove switching frictions	Low	Blur mint, non-parity	1730	-0.0089	-0.0178
Remove switching frictions	Low	Blur mint, tau parity	1079	-0.0085	-0.0169
Remove switching frictions	Low	Blur nonmint, non-parity	509	0.0046	0.0092
Remove switching frictions	Low	Blur nonmint, tau parity	344	0.0070	0.0141

Table 7: Structured Enforcement Counterfactuals by Royalty Group, Enforcement Group, and Mint Structure

Scenario	Royalty Group	Market Structure	n	Δ Mean s_B	Δ Mean Share Gap
Blur matches OpenSea enforcement	Low	Blur stronger, Blur mint	44	0.0000	0.0000
Blur matches OpenSea enforcement	Low	OpenSea stronger, Blur mint	34	0.0403	0.0806
Blur matches OpenSea enforcement	Low	OpenSea stronger, Blur nonmint	3	0.0168	0.0336
Blur matches OpenSea enforcement	Medium	Blur stronger, Blur mint	819	0.0000	0.0000
Blur matches OpenSea enforcement	Medium	Blur stronger, Blur nonmint	96	0.0000	0.0000
Blur matches OpenSea enforcement	Medium	OpenSea stronger, Blur mint	1429	-0.0012	-0.0024
Blur matches OpenSea enforcement	Medium	OpenSea stronger, Blur nonmint	306	-0.0134	-0.0269
Blur matches OpenSea enforcement	High	Blur stronger, Blur mint	112	0.0000	0.0000
Blur matches OpenSea enforcement	High	Blur stronger, Blur nonmint	15	0.0000	0.0000
Blur matches OpenSea enforcement	High	OpenSea stronger, Blur mint	221	-0.0304	-0.0607
Blur matches OpenSea enforcement	High	OpenSea stronger, Blur nonmint	64	-0.0514	-0.1028
Full enforcement both	Low	Blur stronger, Blur mint	44	-0.0093	-0.0187
Full enforcement both	Low	OpenSea stronger, Blur mint	34	0.0403	0.0806
Full enforcement both	Low	OpenSea stronger, Blur nonmint	3	0.0168	0.0336
Full enforcement both	Medium	Blur stronger, Blur mint	819	0.0147	0.0294
Full enforcement both	Medium	Blur stronger, Blur nonmint	96	0.0178	0.0356
Full enforcement both	Medium	OpenSea stronger, Blur mint	1429	-0.0012	-0.0024
Full enforcement both	Medium	OpenSea stronger, Blur nonmint	306	-0.0134	-0.0269
Full enforcement both	High	Blur stronger, Blur mint	112	0.0449	0.0898
Full enforcement both	High	Blur stronger, Blur nonmint	15	0.0514	0.1027
Full enforcement both	High	OpenSea stronger, Blur mint	221	-0.0304	-0.0607
Full enforcement both	High	OpenSea stronger, Blur nonmint	64	-0.0514	-0.1028
OpenSea matches Blur enforcement	Low	Blur stronger, Blur mint	44	-0.0093	-0.0187
OpenSea matches Blur enforcement	Low	OpenSea stronger, Blur mint	34	-0.0000	-0.0000
OpenSea matches Blur enforcement	Low	OpenSea stronger, Blur nonmint	3	0.0000	0.0000
OpenSea matches Blur enforcement	Medium	Blur stronger, Blur mint	819	0.0147	0.0294
OpenSea matches Blur enforcement	Medium	Blur stronger, Blur nonmint	96	0.0178	0.0356
OpenSea matches Blur enforcement	Medium	OpenSea stronger, Blur mint	1429	-0.0000	-0.0000
OpenSea matches Blur enforcement	Medium	OpenSea stronger, Blur nonmint	306	-0.0000	-0.0000
OpenSea matches Blur enforcement	High	Blur stronger, Blur mint	112	0.0449	0.0898
OpenSea matches Blur enforcement	High	Blur stronger, Blur nonmint	15	0.0514	0.1027
OpenSea matches Blur enforcement	High	OpenSea stronger, Blur mint	221	-0.0000	-0.0000
OpenSea matches Blur enforcement	High	OpenSea stronger, Blur nonmint	64	-0.0000	-0.0000
Zero enforcement preference	Low	Blur stronger, Blur mint	44	-0.0134	-0.0268
Zero enforcement preference	Low	OpenSea stronger, Blur mint	34	0.0516	0.1032
Zero enforcement preference	Low	OpenSea stronger, Blur nonmint	3	0.0331	0.0663
Zero enforcement preference	Medium	Blur stronger, Blur mint	819	-0.0226	-0.0452
Zero enforcement preference	Medium	Blur stronger, Blur nonmint	96	-0.0315	-0.0629
Zero enforcement preference	Medium	OpenSea stronger, Blur mint	1429	0.0387	0.0774
Zero enforcement preference	Medium	OpenSea stronger, Blur nonmint	306	0.0509	0.1017
Zero enforcement preference	High	Blur stronger, Blur mint	112	-0.0280	-0.0560
Zero enforcement preference	High	Blur stronger, Blur nonmint	15	-0.0325	-0.0650
Zero enforcement preference	High	OpenSea stronger, Blur mint	221	0.0323	0.0646
Zero enforcement preference	High	OpenSea stronger, Blur nonmint	64	0.0448	0.0896

Table 8: Subgroup Counterfactual Effects

Scenario	Group	n	Δ Mean s_B	Δ Mean Share Gap	Δ Mean Switch
OpenSea matches Blur enforcement	Mint	3808	0.0017	0.0035	-0.0035
OpenSea matches Blur enforcement	Nonmint	3516	0.0014	0.0028	0.0028
OpenSea matches Blur enforcement	High price	3662	0.0020	0.0040	-0.0021
OpenSea matches Blur enforcement	Low price	3662	0.0027	0.0054	-0.0028
OpenSea matches Blur enforcement	Tau parity	3581	0.0003	0.0006	-0.0006
OpenSea matches Blur enforcement	Kappa parity	4172	0.0047	0.0094	-0.0012
Full enforcement both	Mint	3808	0.0003	0.0007	-0.0021
Full enforcement both	Nonmint	3516	0.0014	0.0028	0.0018
Full enforcement both	High price	3662	-0.0017	-0.0033	-0.0005
Full enforcement both	Low price	3662	0.0034	0.0069	-0.0041
Full enforcement both	Tau parity	3581	-0.0004	-0.0007	-0.0007
Full enforcement both	Kappa parity	4172	0.0047	0.0094	-0.0012
Remove switching frictions	Mint	3808	-0.0873	-0.1746	0.0873
Remove switching frictions	Nonmint	3516	0.0291	0.0583	0.0291
Remove switching frictions	High price	3662	-0.0615	-0.1229	0.0802
Remove switching frictions	Low price	3662	-0.0010	-0.0021	0.0024
Remove switching frictions	Tau parity	3581	-0.0107	-0.0213	0.0769
Remove switching frictions	Kappa parity	4172	-0.0577	-0.1154	0.0804
Blur matches OpenSea enforcement	Mint	3808	-0.0014	-0.0028	0.0000
Blur matches OpenSea enforcement	Nonmint	3516	-0.0000	-0.0000	0.0003
Blur matches OpenSea enforcement	High price	3662	-0.0032	-0.0065	0.0009
Blur matches OpenSea enforcement	Low price	3662	-0.0015	-0.0030	-0.0008
Blur matches OpenSea enforcement	Tau parity	3581	-0.0011	-0.0022	0.0001
Blur matches OpenSea enforcement	Kappa parity	4172	-0.0015	-0.0030	0.0000
Zero enforcement preference	Mint	3808	0.0033	0.0066	-0.0044
Zero enforcement preference	Nonmint	3516	0.0119	0.0237	-0.0024
Zero enforcement preference	High price	3662	0.0054	0.0109	-0.0024
Zero enforcement preference	Low price	3662	0.0097	0.0193	-0.0042
Zero enforcement preference	Tau parity	3581	0.0000	0.0001	-0.0006
Zero enforcement preference	Kappa parity	4172	0.0023	0.0046	-0.0004

Table 9: Counterfactual Outcomes by Royalty Bin

Scenario	Royalty Bin	n	Base Gap	CF Gap	Base s_B	CF s_B	Base Switch	CF Switch	Δ Gap	Δs_B	Δ Switch
opensea matches blur enforcement	[0.0, 1.0)	126	0.6690	0.6630	0.8345	0.8315	0.1722	0.1752	-0.0061	-0.0030	0.0029
opensea matches blur enforcement	[1.0, 2.5)	52	0.4425	0.4389	0.7213	0.7195	0.3547	0.3565	-0.0036	-0.0018	0.0018
opensea matches blur enforcement	[2.5, 5.0)	1178	0.6513	0.6523	0.8256	0.8261	0.2116	0.2112	0.0010	0.0005	-0.0004
opensea matches blur enforcement	[5.0, 7.5)	4933	0.5419	0.5488	0.7710	0.7744	0.2840	0.2815	0.0069	0.0034	-0.0025
opensea matches blur enforcement	[7.5, 10.0)	692	0.5367	0.5499	0.7684	0.7749	0.2839	0.2790	0.0131	0.0066	-0.0050
opensea matches blur enforcement	[10.0, 15.0)	343	0.5248	0.5425	0.7624	0.7713	0.3458	0.3398	0.0177	0.0088	-0.0060
full enforcement both	[0.0, 1.0)	126	0.6690	0.6718	0.8345	0.8359	0.1722	0.1709	0.0027	0.0014	-0.0013
full enforcement both	[1.0, 2.5)	52	0.4425	0.4718	0.7213	0.7359	0.3547	0.3422	0.0293	0.0147	-0.0125
full enforcement both	[2.5, 5.0)	1178	0.6513	0.6579	0.8256	0.8289	0.2116	0.2086	0.0066	0.0033	-0.0030
full enforcement both	[5.0, 7.5)	4933	0.5419	0.5432	0.7710	0.7716	0.2840	0.2822	0.0013	0.0007	-0.0019
full enforcement both	[7.5, 10.0)	692	0.5367	0.5339	0.7684	0.7670	0.2839	0.2835	-0.0028	-0.0014	-0.0005
full enforcement both	[10.0, 15.0)	343	0.5248	0.5065	0.7624	0.7533	0.3458	0.3424	-0.0183	-0.0092	-0.0034
remove switching frictions	[0.0, 1.0)	126	0.6690	0.4298	0.8345	0.7149	0.1722	0.2951	-0.2393	-0.1196	0.1229
remove switching frictions	[1.0, 2.5)	52	0.4425	0.4144	0.7213	0.7072	0.3547	0.3688	-0.0281	-0.0140	0.0140
remove switching frictions	[2.5, 5.0)	1178	0.6513	0.4293	0.8256	0.7146	0.2116	0.3250	-0.2220	-0.1110	0.1134
remove switching frictions	[5.0, 7.5)	4933	0.5419	0.4429	0.7710	0.7214	0.2840	0.3613	-0.0990	-0.0495	0.0772
remove switching frictions	[7.5, 10.0)	692	0.5367	0.4481	0.7684	0.7240	0.2839	0.3404	-0.0887	-0.0443	0.0565
remove switching frictions	[10.0, 15.0)	343	0.5248	0.4596	0.7624	0.7298	0.3458	0.3947	-0.0652	-0.0326	0.0488
blur matches opensea enforcement	[0.0, 1.0)	126	0.6690	0.6778	0.8345	0.8389	0.1722	0.1679	0.0088	0.0044	-0.0043
blur matches opensea enforcement	[1.0, 2.5)	52	0.4425	0.4754	0.7213	0.7377	0.3547	0.3404	0.0329	0.0165	-0.0143
blur matches opensea enforcement	[2.5, 5.0)	1178	0.6513	0.6569	0.8256	0.8284	0.2116	0.2089	0.0056	0.0028	-0.0026
blur matches opensea enforcement	[5.0, 7.5)	4933	0.5419	0.5363	0.7710	0.7682	0.2840	0.2847	-0.0056	-0.0028	0.0007
blur matches opensea enforcement	[7.5, 10.0)	692	0.5367	0.5208	0.7684	0.7604	0.2839	0.2884	-0.0160	-0.0080	0.0045
blur matches opensea enforcement	[10.0, 15.0)	343	0.5248	0.4888	0.7624	0.7444	0.3458	0.3484	-0.0360	-0.0180	0.0026
zero enforcement preference	[0.0, 1.0)	126	0.6690	0.6714	0.8345	0.8357	0.1722	0.1711	0.0024	0.0012	-0.0011
zero enforcement preference	[1.0, 2.5)	52	0.4425	0.4834	0.7213	0.7417	0.3547	0.3388	0.0409	0.0204	-0.0159
zero enforcement preference	[2.5, 5.0)	1178	0.6513	0.6649	0.8256	0.8324	0.2116	0.2062	0.0136	0.0068	-0.0054
zero enforcement preference	[5.0, 7.5)	4933	0.5419	0.5568	0.7710	0.7784	0.2840	0.2809	0.0149	0.0075	-0.0031
zero enforcement preference	[7.5, 10.0)	692	0.5367	0.5469	0.7684	0.7735	0.2839	0.2820	0.0102	0.0051	-0.0019
zero enforcement preference	[10.0, 15.0)	343	0.5248	0.5393	0.7624	0.7697	0.3458	0.3463	0.0145	0.0072	0.0004
targeted full enforcement high royalty	[0.0, 1.0)	126	0.6690	0.6690	0.8345	0.8345	0.1722	0.1722	0.0000	0.0000	0.0000
targeted full enforcement high royalty	[1.0, 2.5)	52	0.4425	0.4425	0.7213	0.7213	0.3547	0.3547	0.0000	0.0000	0.0000
targeted full enforcement high royalty	[2.5, 5.0)	1178	0.6513	0.6513	0.8256	0.8256	0.2116	0.2116	0.0000	0.0000	0.0000
targeted full enforcement high royalty	[5.0, 7.5)	4933	0.5419	0.5419	0.7710	0.7710	0.2840	0.2840	0.0000	0.0000	0.0000
targeted full enforcement high royalty	[7.5, 10.0)	692	0.5367	0.5339	0.7684	0.7670	0.2839	0.2835	-0.0028	-0.0014	-0.0005
targeted full enforcement high royalty	[10.0, 15.0)	343	0.5248	0.5065	0.7624	0.7533	0.3458	0.3424	-0.0183	-0.0092	-0.0034

Notes: Base columns refer to the baseline model prediction, CF columns refer to the counterfactual prediction, and the Δ columns are the counterfactual minus baseline values.